ArchR-based analysis of chromatin accessibility, transcription-factor activity and regulatory elements

Sakshi Parate — M.Sc. Bioinformatics, Saarland University

This report presents the complete single-cell ATAC-seq analysis performed using ArchR, including preprocessing and quality control, peak calling, dimensionality reduction, clustering, gene activity, transcription-factor motif analysis, integration with scRNA-seq data, differential accessibility, co-accessibility and potential enhancer identification.

1. Analysis Overview

Single-cell ATAC-seq (scATAC-seq) was analyzed using an ArchR-based workflow to characterize chromatin accessibility at single-cell resolution. The analysis proceeded from fragment processing and quality control through peak calling, dimensionality reduction, clustering, gene activity, transcription-factor motif analysis, integration with gene-expression data, differential accessibility and regulatory-element analysis.

Analysis workflow

Fragment files → Arrow files → Quality control and filtering → Doublet detection → Pseudo-bulk peak calling → Joint peak set and PeakMatrix → Iterative LSI → UMAP and Harmony batch correction → Louvain clustering → GeneScoreMatrix and marker genes → MAGIC smoothing → TF motif annotation and chromVAR activity → scRNA-seq integration → GeneIntegrationMatrix → Correlation and cluster-label comparison → Peak-gene linkage → Differential peak accessibility → TF motif enrichment → TF footprinting → Co-accessibility → Potential enhancer identification.

Software and methods

- R
- ArchR
- chromVAR
- Harmony
- MACS2
- cisBP transcription-factor motif annotations

2. Data Preparation and Quality Control

2.1 Data preparation

The required fragment datasets were downloaded and prepared for the analysis pipeline. The reference genome was set to hg38, with parallel processing enabled where supported.

2.2 Fragment import and initial filtering

The downloaded fragment files were imported into ArchR Arrow files. Initial lenient filtering used filterFrags = 550 and filterTSS = 5 to retain cells with sufficient fragment coverage and transcription start site enrichment for downstream processing.

2.3 Doublet detection

Doublet identification was performed and predicted doublets were removed from the dataset. Doublet removal was followed by stricter filtering based on fragment counts and TSS enrichment before the final analysis set was established.

2.4 Joint ArchR project

All samples were combined into a joint ArchR project containing 9,460 cells before the final filtering steps. The median TSS enrichment was 9.56 and the median number of fragments was 5,577. The initial tile matrix contained 6,062,095 tiles across 9,460 cells.

2.5 Cells per sample

Sample	Cells
ATAC_555_2	1,227
ATAC_557	851
ATAC_EV08	970
ATAC_HIP02_frozen	6,412
Total	9,460

2.6 Fragment length distribution

Fragment length distributions were examined across all samples. The characteristic periodicity is consistent with nucleosome organization: fragments shorter than 100 bp mainly represent nucleosome-free accessible chromatin, the approximately 200 bp peak represents mononucleosomal DNA protected by one nucleosome, and the approximately 400 bp peak represents dinucleosomal DNA protected by two adjacent nucleosomes. The observed periodic pattern is consistent with preserved nucleosome organization and good library quality.

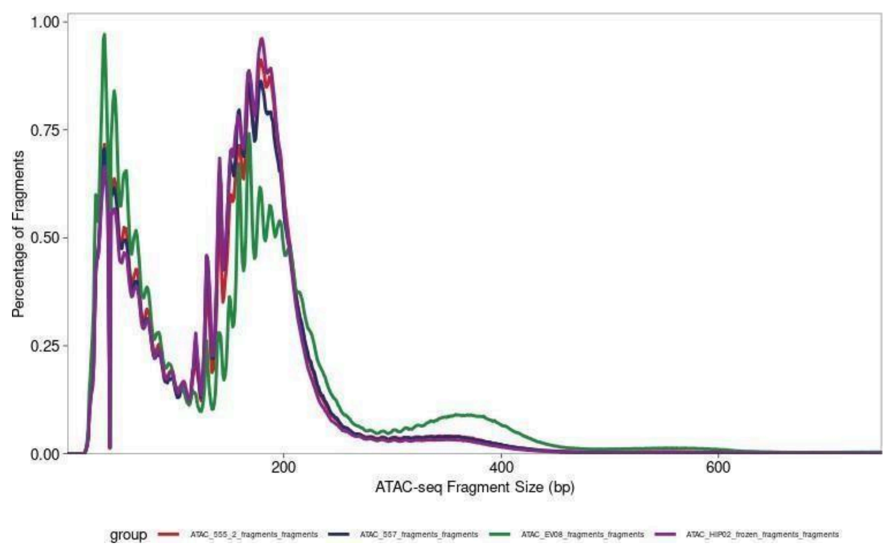


Figure 1. Fragment length distribution across all ATAC-seq samples.

2.7 TSS enrichment

TSS enrichment profiles were examined for the samples. The profiles show a pronounced enrichment around transcription start sites, consistent with preferential accessibility at active regulatory regions.

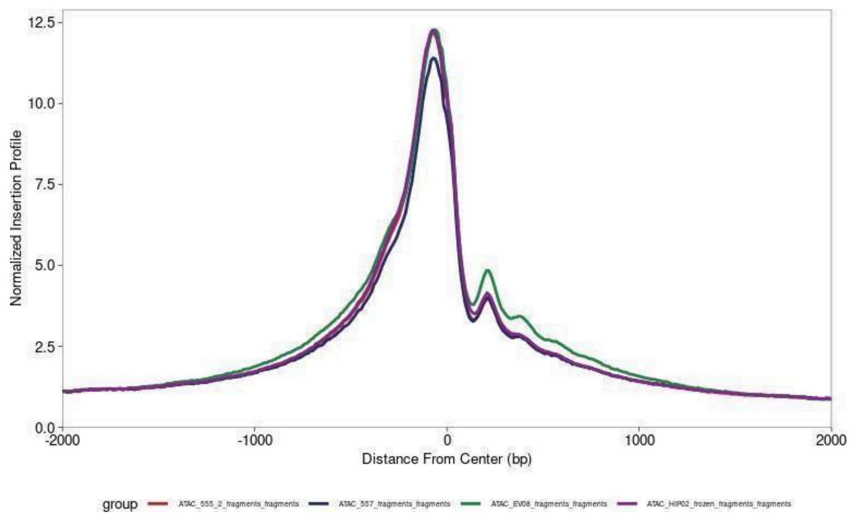


Figure 2. TSS enrichment profiles across the four samples.

2.8 Fragment count versus TSS enrichment

The relationship between the number of fragments and TSS enrichment was examined separately for each sample. The distributions were broadly consistent across samples, with no sample showing a clear low-quality pattern.

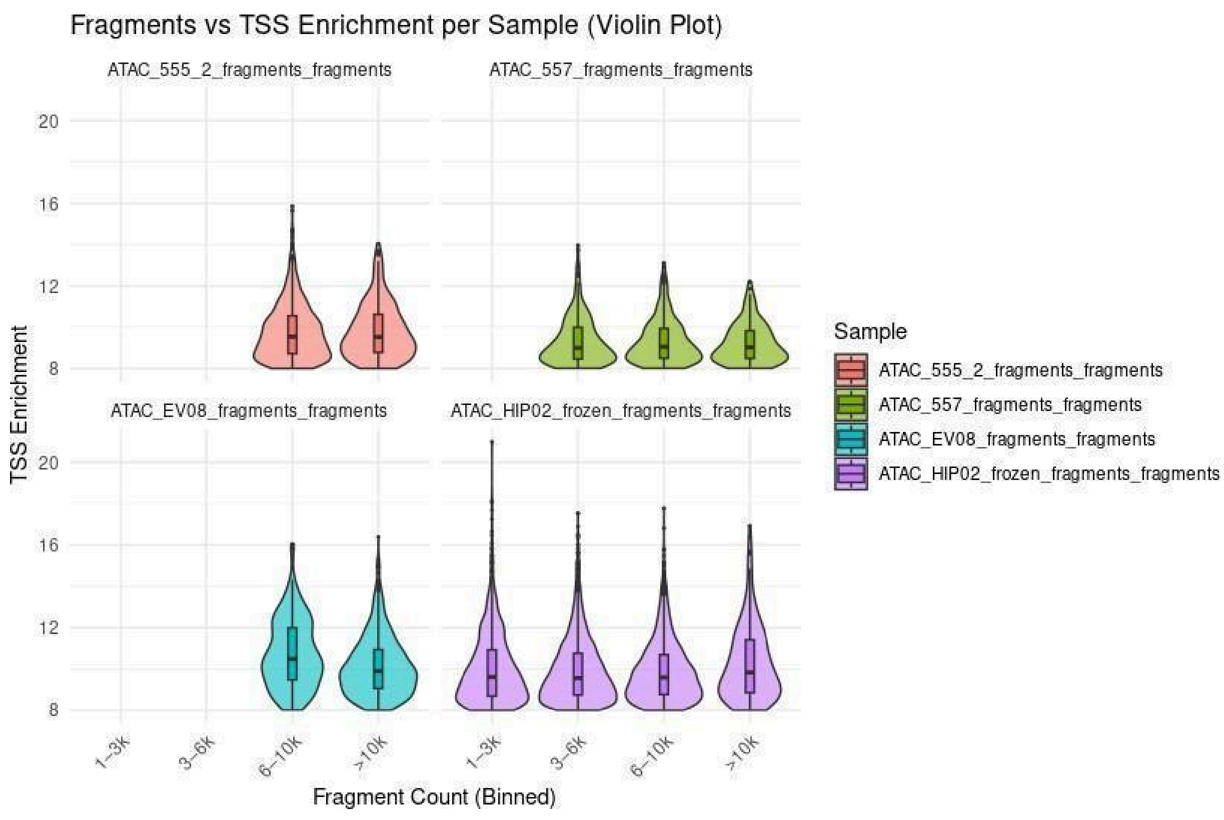


Figure 3. Distribution of TSS enrichment across fragment-count bins for the four samples.

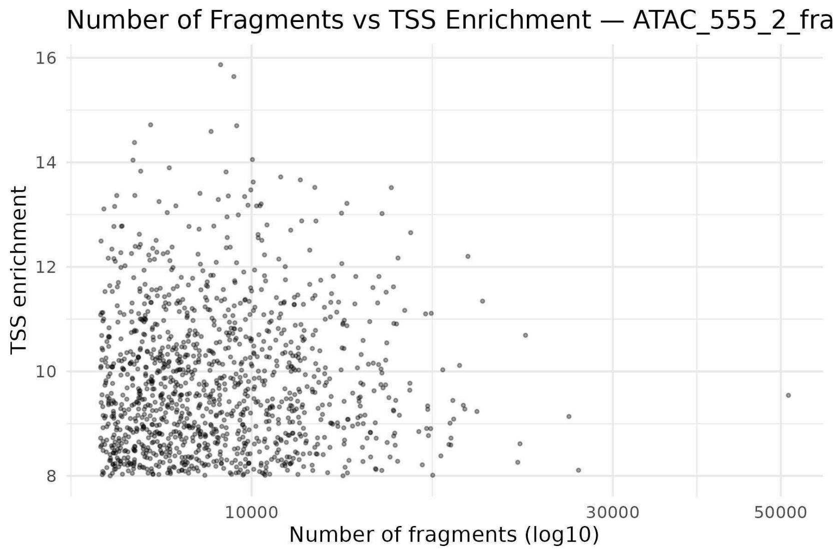


Figure 3a. Fragment count versus TSS enrichment for ATAC_555_2.

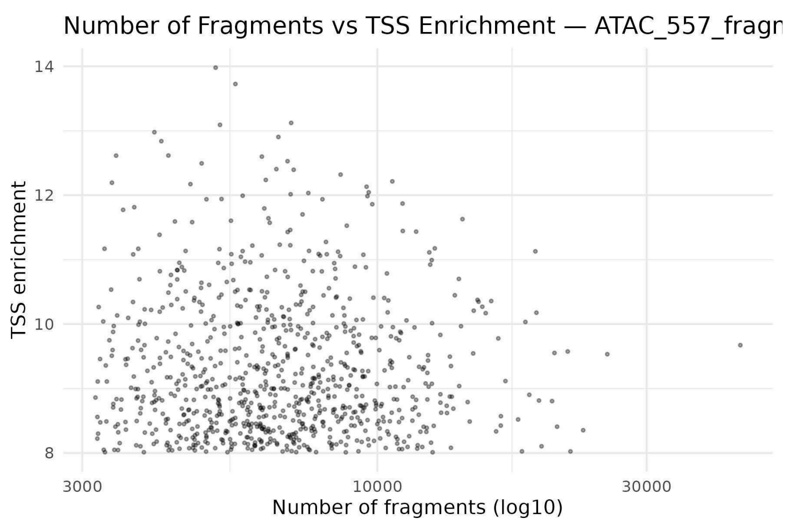


Figure 3b. Fragment count versus TSS enrichment for ATAC_557.

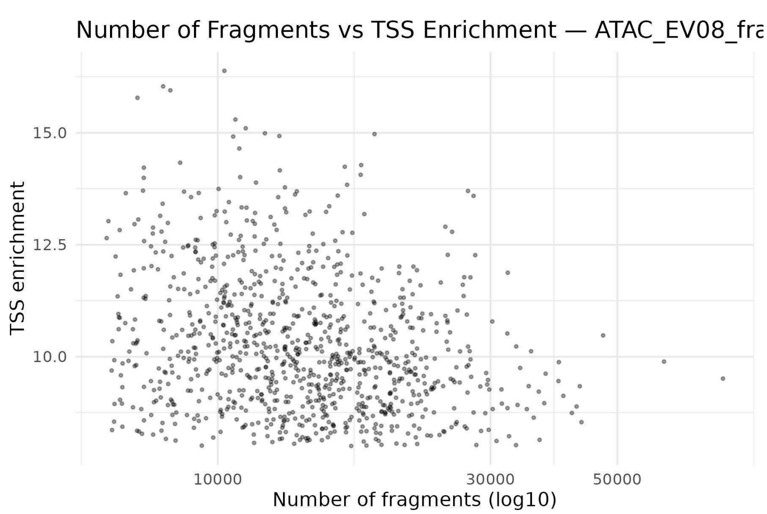


Figure 3c. Fragment count versus TSS enrichment for ATAC_EV08.

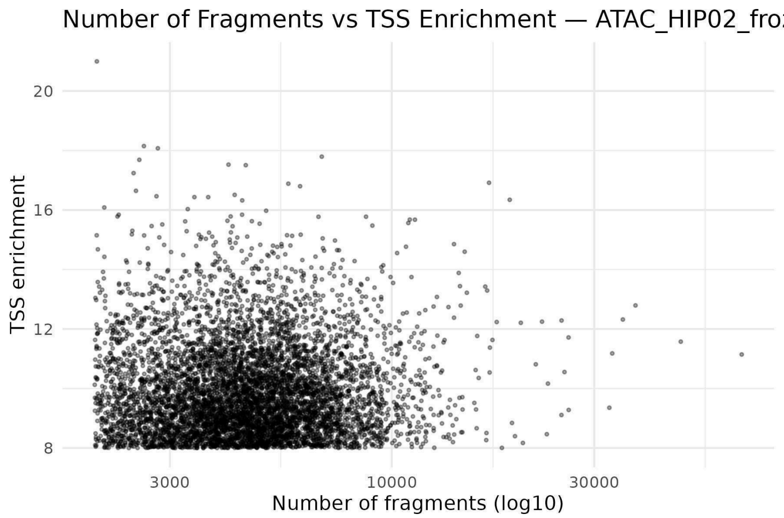


Figure 3d. Fragment count versus TSS enrichment for ATAC_HIP02_frozen.

No samples were identified as clear outliers or low-quality samples. Fragment-length periodicity, TSS enrichment profiles and fragment-count/TSS relationships were consistent across samples. ATAC_HIP02_frozen contains many more cells, so its scatter plot contains more points, but it follows the same overall quality pattern.

2.9 Final filtering

Stricter filtering was applied using fragment counts, TSS enrichment and doublet removal. The final filtered ArchR project contained 5,089 cells, with a median TSS enrichment of 9.525 and a median fragment count of 5,108. The filtered tile matrix contained 6,062,095 tiles across 5,089 cells.

3. Peak Calling and Peak Analysis

3.1 Peak calling

Peak calling was performed on the strictly filtered ArchR project after low-quality cells and predicted doublets had been removed. Cells were grouped to generate pseudo-bulk replicates, which aggregate accessibility signal and provide greater read depth than individual single-cell profiles.

Pseudo-bulk peak calling was used because single-cell ATAC-seq profiles are sparse and individually provide insufficient read depth for reliable peak detection. Calling peaks only on the complete merged dataset can also favor abundant cell populations and mask regulatory elements specific to rarer populations. Pseudo-bulking preserves biological heterogeneity while increasing signal-to-noise, helping detect regulatory elements represented in smaller populations.

3.2 Joint reproducible peak set and PeakMatrix

A joint reproducible peak set was computed from the pseudo-bulk groups. The resulting peak set was added to the ArchR project, and a peak-by-cell accessibility matrix (PeakMatrix) was generated for downstream analyses.

3.3 Cluster marker peaks

Differential accessibility of peaks between clusters was tested using the PeakMatrix.

Parameter	Setting / purpose
useMatrix	PeakMatrix — uses peak-level accessibility values.
groupBy	Clusters — compares accessibility between cell clusters.

testMethod	Wilcoxon — non-parametric test suitable for sparse single-cell ATAC-seq data.
bias	TSSEnrichment and log10(nFrags) — corrects for chromatin-signal quality and sequencing-depth effects.
FDR cutoff	≤ 0.01 — controls the false discovery rate.
Log2FC cutoff	≥ 1 — retains peaks with biologically meaningful accessibility differences.

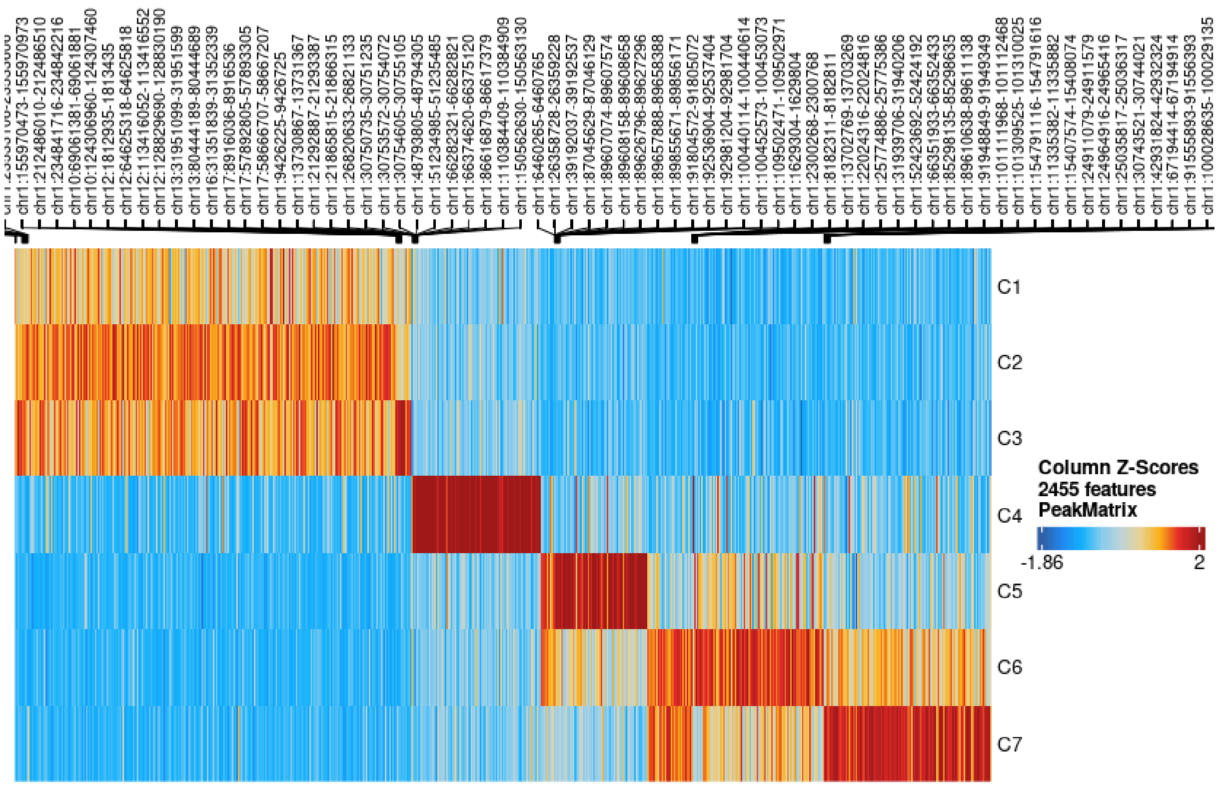


Figure 4. Heatmap of accessibility in cluster marker peaks. The displayed heatmap contains 2,455 marker-peak features across clusters C1–C7.

3.4 Gene-region accessibility profiles

Accessibility profiles were examined around CD8A, CD14, GATA1, PAX5 and TBX21 to visualize cluster-specific chromatin accessibility and the relationship between accessible regions, called peaks and gene annotations.

Figure 5. Genome-region accessibility profile around CD8A.

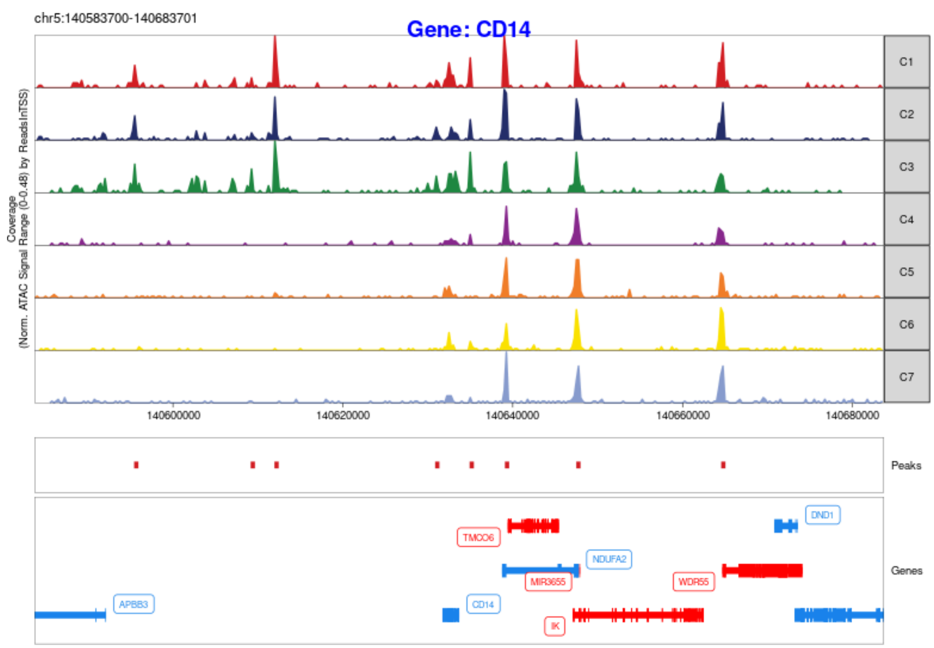


Figure 5. Genome-region accessibility profile around CD14.

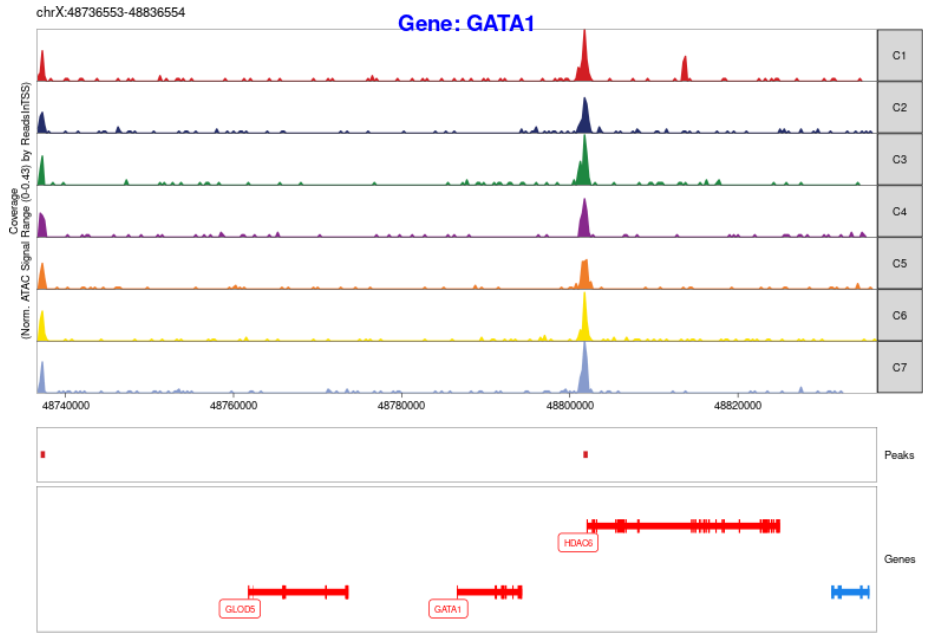


Figure 5. Genome-region accessibility profile around GATA1.

Figure 5. Genome-region accessibility profile around PAX5.

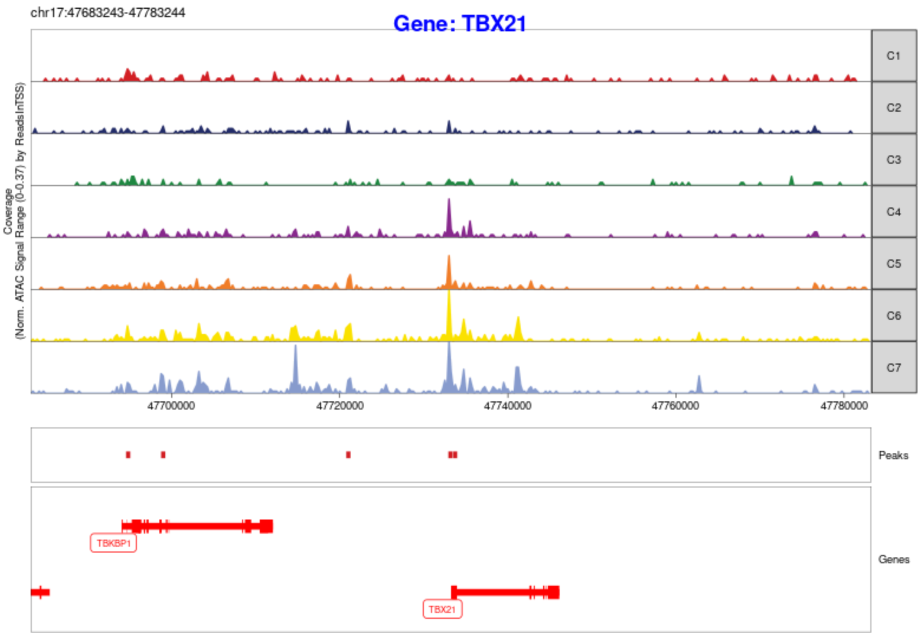


Figure 5. Genome-region accessibility profile around TBX21.

4. Dimensionality Reduction, Batch Correction and Clustering

4.1 Iterative LSI

Iterative LSI was used for dimensionality reduction of the scATAC-seq accessibility matrix. scATAC-seq data are highly sparse and largely binary-like, because most peaks are inaccessible in most cells. PCA is designed for continuous data and therefore does not model this structure as effectively. LSI is well suited to sparse count-like matrices and reduces the influence of sequencing depth and technical noise while capturing meaningful chromatin-accessibility patterns.

4.2 Exclusion of LSI component 1

LSI component 1 was excluded from clustering and UMAP generation because it was strongly correlated with sequencing depth, measured by the number of fragments. This indicates that LSI 1 primarily captured technical variation. Including it would risk separating cells according to read depth rather than biological identity.

4.3 UMAP coloured by sample and QC metrics

UMAP was generated from the LSI representation and visualized using sample identity, TSS enrichment and number of fragments.

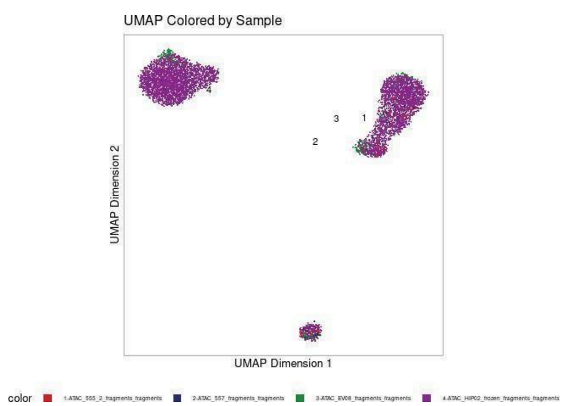


Figure 6a. UMAP coloured by sample identity before batch correction.

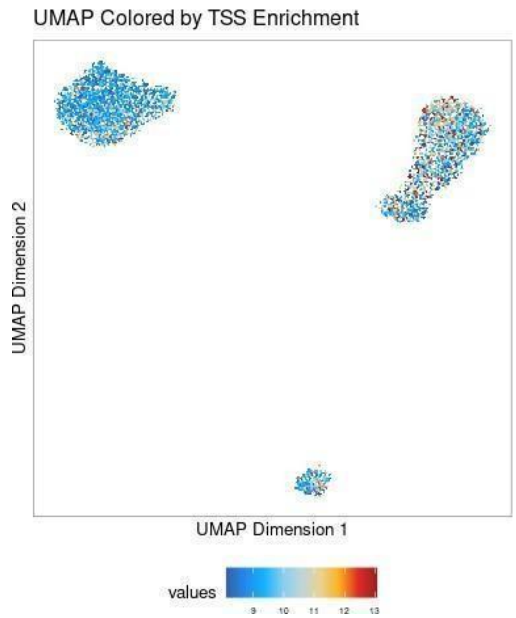


Figure 6b. UMAP coloured by TSS enrichment.

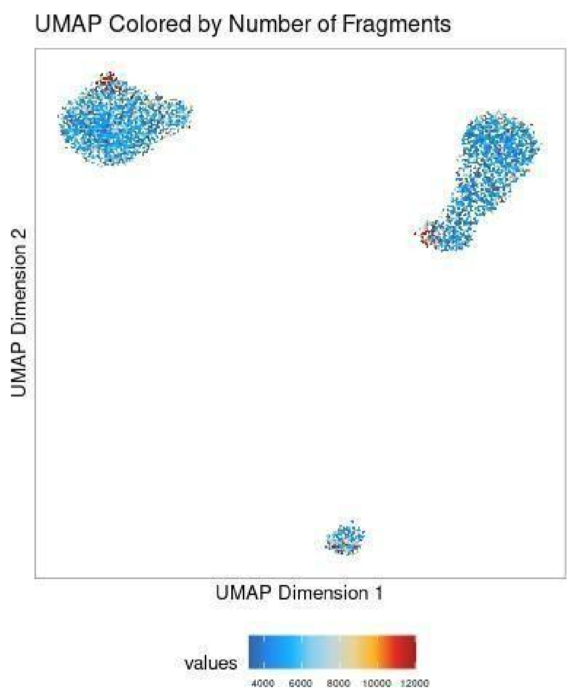


Figure 6c. UMAP coloured by number of fragments.

4.4 Batch effects and Harmony correction

Before correction, cells from different samples formed distinct, largely non-overlapping regions in the UMAP, indicating strong sample-associated technical variation. Harmony batch correction was applied to reduce technical differences and align cells in a shared latent space.

After Harmony correction, mixing between samples improved while the overall biological structure was retained. The corrected UMAP therefore provided a more suitable representation for downstream clustering.

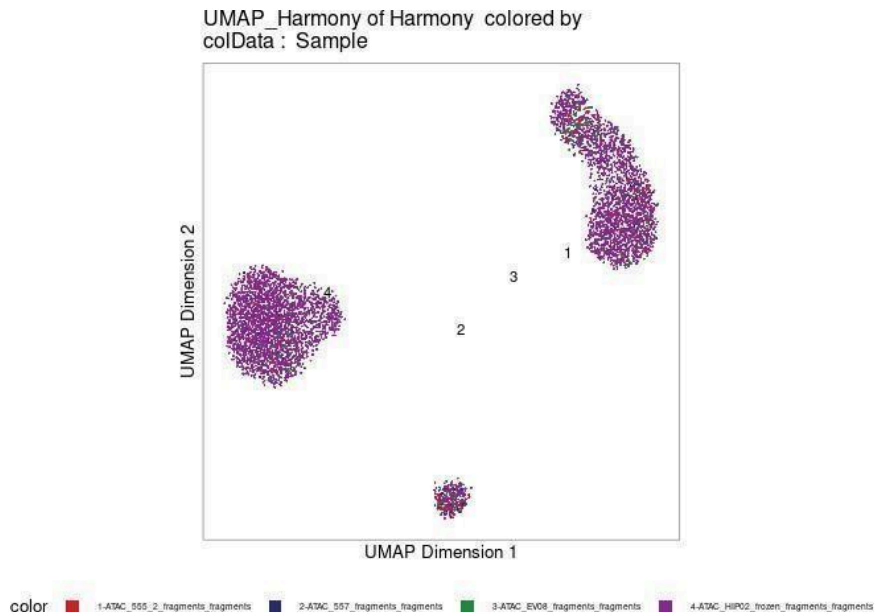


Figure 7a. Harmony-corrected UMAP coloured by sample.

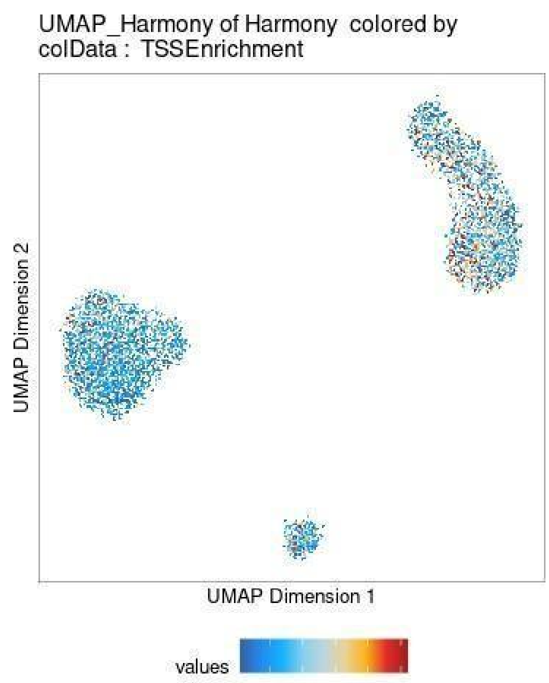


Figure 7b. Harmony-corrected UMAP coloured by TSS enrichment.

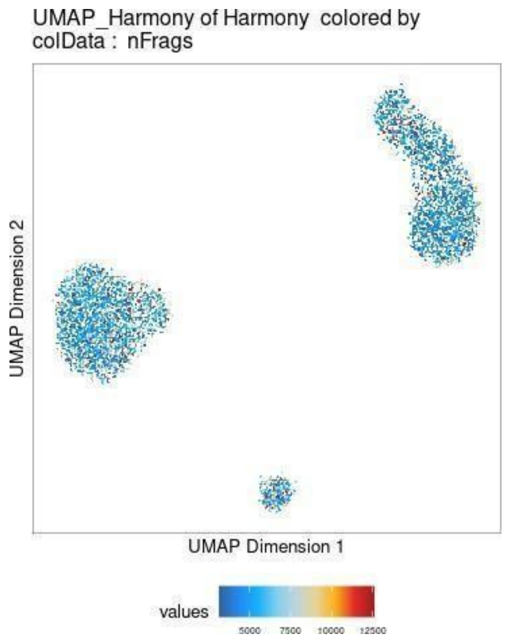


Figure 7c. Harmony-corrected UMAP coloured by number of fragments.

4.5 Louvain clustering

Louvain clustering was applied to the final cell representation. Seven clusters (C1–C7) were identified.

Cluster	Number of cells
C1	335
C2	1,894
C3	279
C4	318
C5	402
C6	786
C7	1,075
Total	5,089

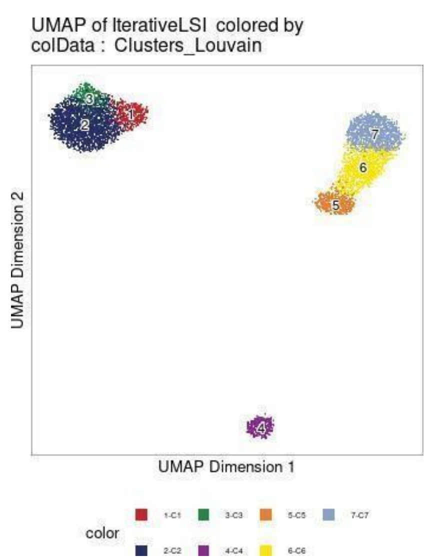


Figure 8. UMAP representation coloured by Louvain cluster.

4.6 Sample composition of clusters

The majority of cells in most clusters originated from ATAC_HIP02_frozen. C1, C2, C6 and C7 were especially dominated by this sample, while C3 contained a larger contribution from ATAC_EV08. Cluster C4 showed the most balanced contribution from the four samples among the major clusters.

Cluster	ATAC_555_2	ATAC_557	ATAC_EV08	ATAC_HIP02_frozen
C1	1.19%	0.00%	0.30%	98.51%
C2	2.64%	0.63%	0.42%	96.29%
C3	2.51%	1.08%	24.73%	71.94%
C4	17.30%	12.27%	16.04%	54.38%
C5	13.94%	1.99%	18.41%	65.67%
C6	7.25%	2.93%	1.66%	88.15%
C7	1.67%	0.47%	3.91%	93.96%

5. Gene Activity and Marker Genes

5.1 Gene activity scores

Gene activity scores were computed by aggregating chromatin-accessibility signal from promoter regions and nearby regulatory elements for each gene. The resulting GeneScoreMatrix contained 24,919 genes across 5,089 filtered cells.

5.2 Cluster marker genes

Cluster-specific marker genes were identified using differential gene-activity analysis based on the GeneScoreMatrix. A Wilcoxon rank-sum test was used, with technical biases from TSS enrichment and log10(nFrags) accounted for.

Parameter	Setting
Statistical test	Wilcoxon rank-sum test
Bias correction	TSSEnrichment and log10(nFrags)
FDR cutoff	≤ 0.05
Log2FC cutoff	≥ 1

These thresholds were used to retain marker genes that were statistically significant and showed a biologically meaningful increase in gene activity.

Cluster	Representative marker genes
C1	TM4SF5, ZMYND15, MRVI1, ZFAND5, IFI30
C2	TTC7A, ASGR2, ANPEP, MIR3945HG, MIR3945
C3	LINC00656, NLRP3, SLC43A2, RIN2, RIPK2
C4	BLK, CCR6, PAX5, HLA-DQA1, EBF1
C5	LEF1, BCL11B, IL7R, IGSF9B, LRRC8C-DT
C6	BCL11B, CD8A, FAM71B, ITK, LINC02520
C7	CD247, TARP, SPON2, KLRD1, GZMB

5.3 Marker-gene visualization with and without MAGIC smoothing

MAGIC (Markov Affinity-based Graph Imputation of Cells) was used as a graph-based smoothing method to propagate gene-activity information between cells with similar profiles. UMAPs were inspected before and after smoothing to compare the representation of marker-gene activity.

Representative GeneScoreMatrix UMAPs for cluster marker genes are shown below.

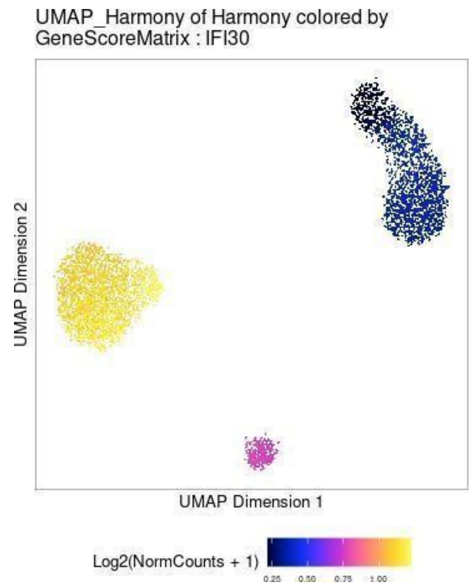


Figure 9. GeneScoreMatrix UMAP for IFI30 without MAGIC smoothing.

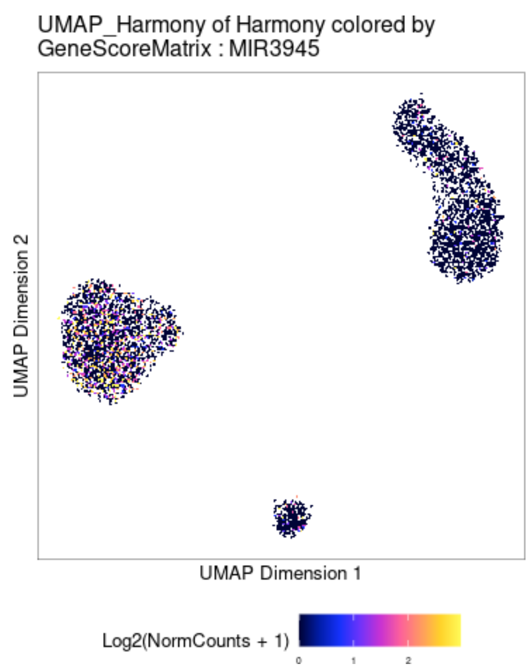


Figure 9. GeneScoreMatrix UMAP for MIR3945 without MAGIC smoothing.

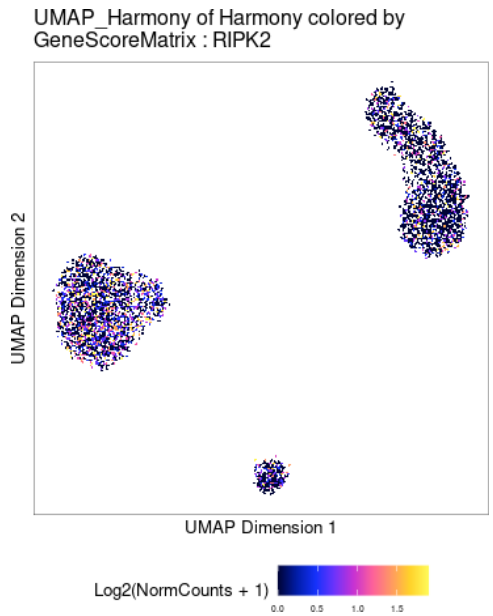


Figure 9. GeneScoreMatrix UMAP for RIPK2 without MAGIC smoothing.

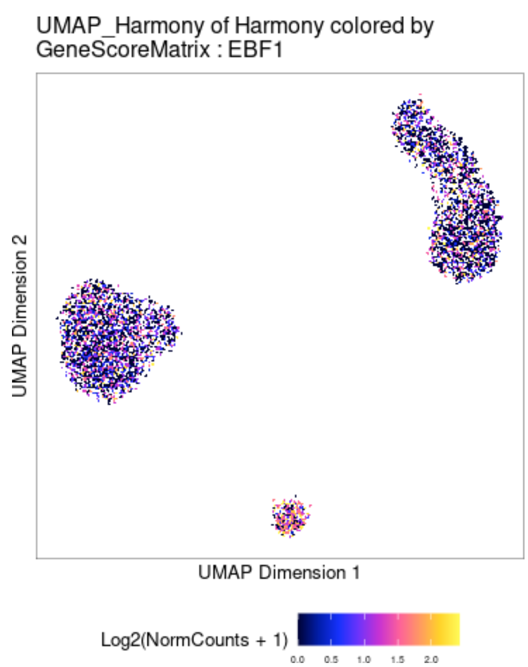


Figure 9. GeneScoreMatrix UMAP for EBF1 without MAGIC smoothing.

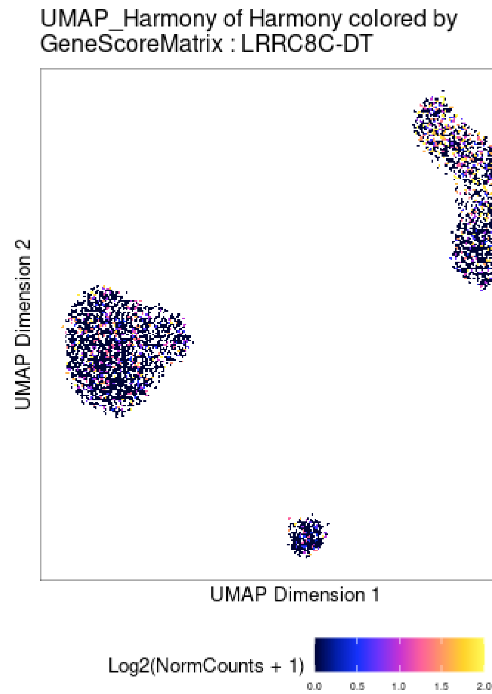


Figure 9. GeneScoreMatrix UMAP for LRRC8C-DT without MAGIC smoothing.

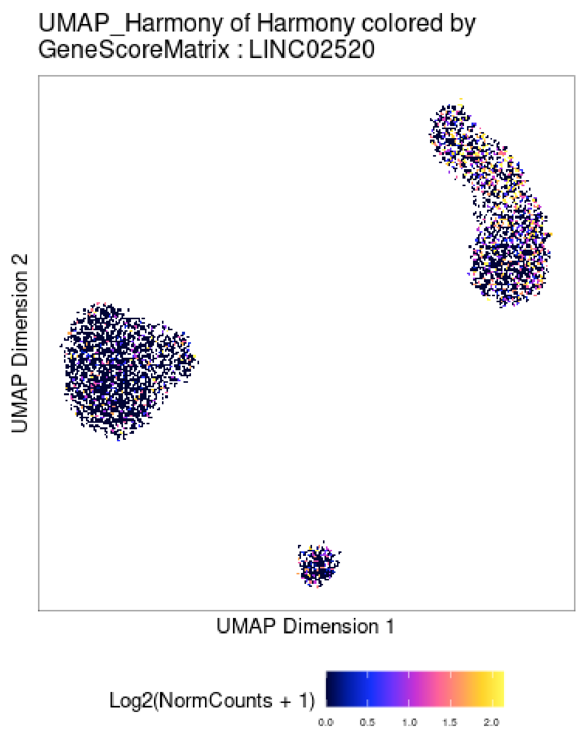


Figure 9. GeneScoreMatrix UMAP for LINC02520 without MAGIC smoothing.

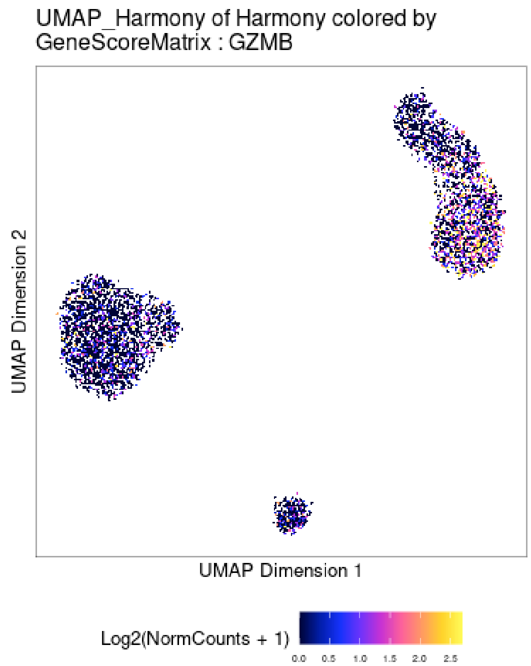


Figure 9. GeneScoreMatrix UMAP for GZMB without MAGIC smoothing.

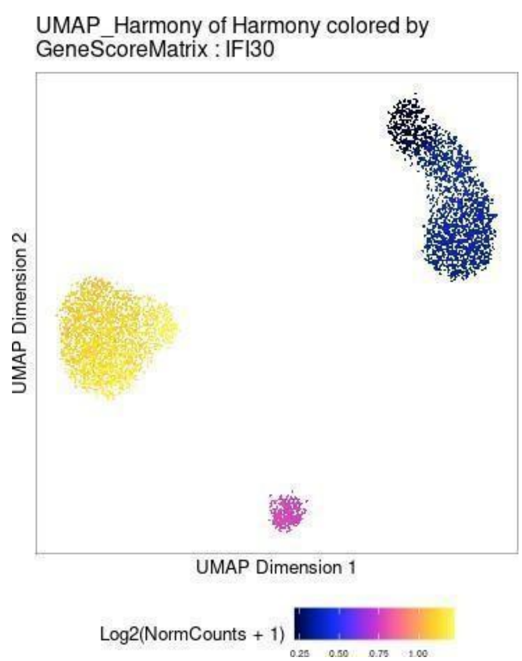


Figure 10. GeneScoreMatrix UMAP for IFI30 with MAGIC smoothing, as presented in the analysis output.

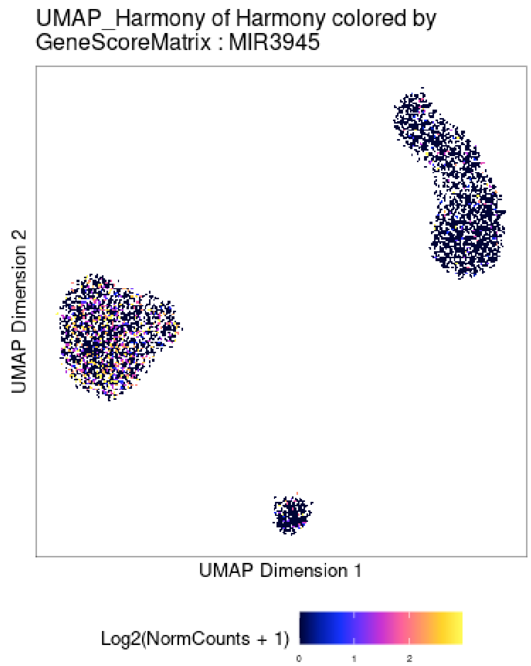


Figure 10. GeneScoreMatrix UMAP for MIR3945 with MAGIC smoothing, as presented in the analysis output.

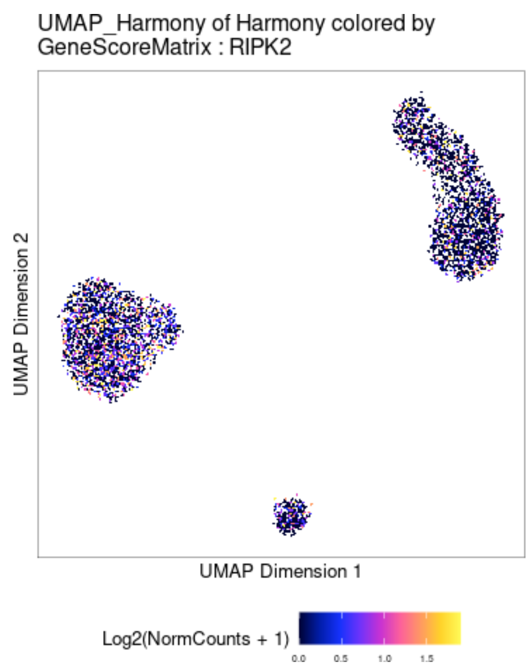


Figure 10. GeneScoreMatrix UMAP for RIPK2 with MAGIC smoothing, as presented in the analysis output.

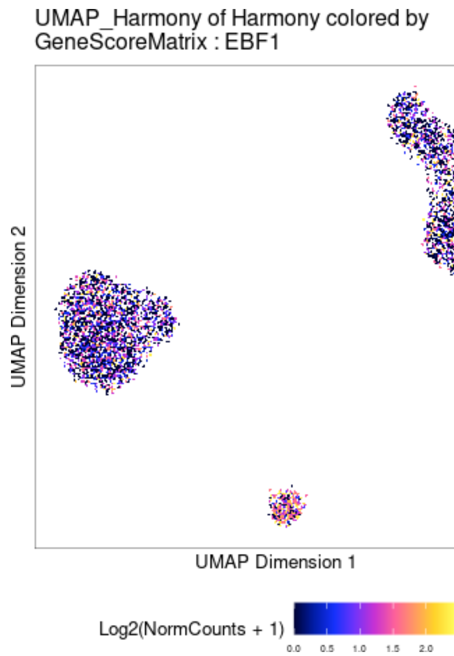


Figure 10. GeneScoreMatrix UMAP for EBF1 with MAGIC smoothing, as presented in the analysis output.

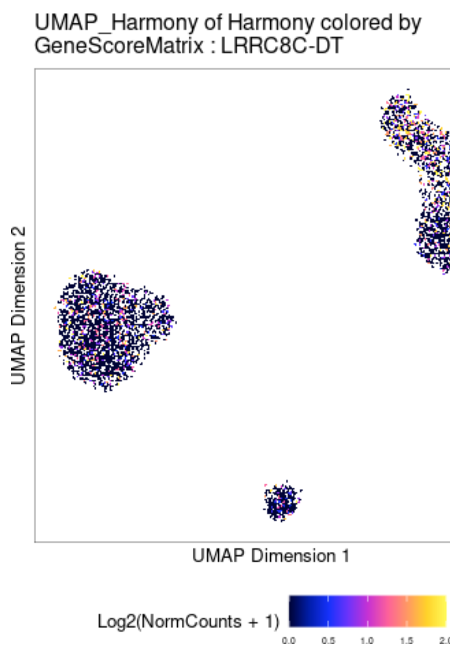


Figure 10. GeneScoreMatrix UMAP for LRRC8C-DT with MAGIC smoothing, as presented in the analysis output.

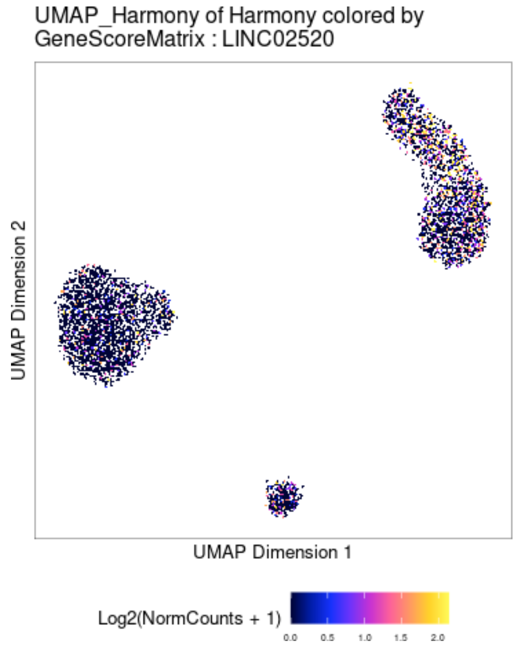


Figure 10. GeneScoreMatrix UMAP for LINC02520 with MAGIC smoothing, as presented in the analysis output.

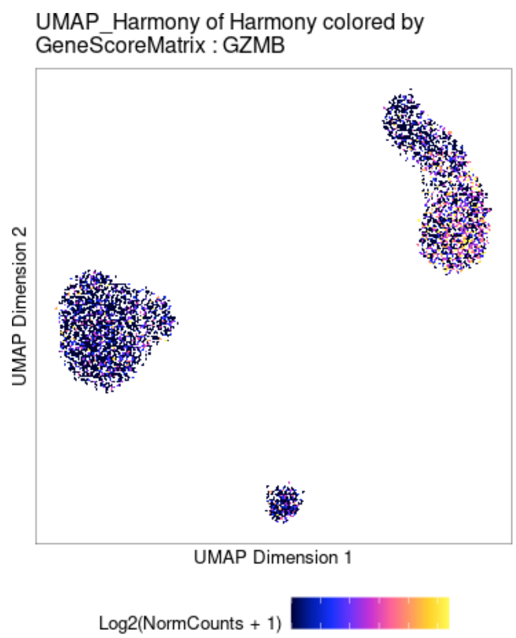


Figure 10. GeneScoreMatrix UMAP for GZMB with MAGIC smoothing, as presented in the analysis output.

The smoothed representations are intended to make gene-activity patterns more continuous across neighbouring cells, while the unsmoothed views retain the original cell-level signal.

5.4 Main idea behind MAGIC

MAGIC uses a cell-cell affinity graph to smooth sparse single-cell measurements. Information from cells with similar profiles is propagated to neighbouring cells, helping reveal biological patterns that can be obscured by sparse measurements. Smoothing should be interpreted as an imputation-based visualization aid rather than direct measurement of gene activity.

6. Transcription-Factor Motif Activity

6.1 Motif annotation

The cisBP transcription-factor motif annotation database was selected because it provides a comprehensive collection of curated transcription-factor binding motifs suitable for scATAC-seq analysis.

Motif annotations were added to the ArchR project using addMotifAnnotations() with the cisBP motif set. Background peaks were generated with addBgdPeaks(), and motif deviation scores were computed using addDeviationsMatrix(). These deviation scores provide a single-cell measure of accessibility associated with TF motifs.

6.2 Most variable TF motifs

The most variable TF motifs, based on combined variability scores, were SPIB_336, SPI1_322, SPIC_344, FOS_137, JUNB_139 and CEBPA_155. The two most variable motifs were SPIB_336 and SPI1_322.

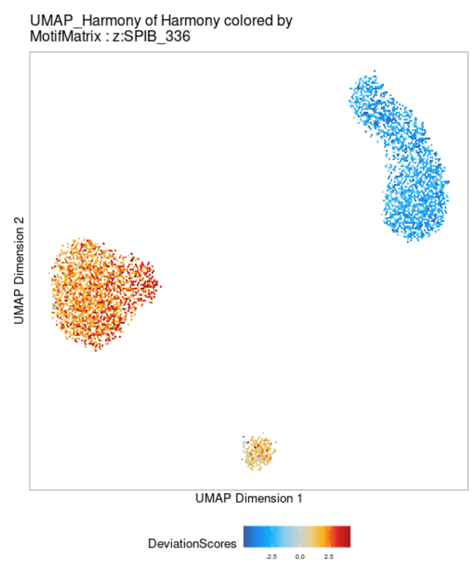


Figure 11. UMAP showing SPIB_336 motif activity.

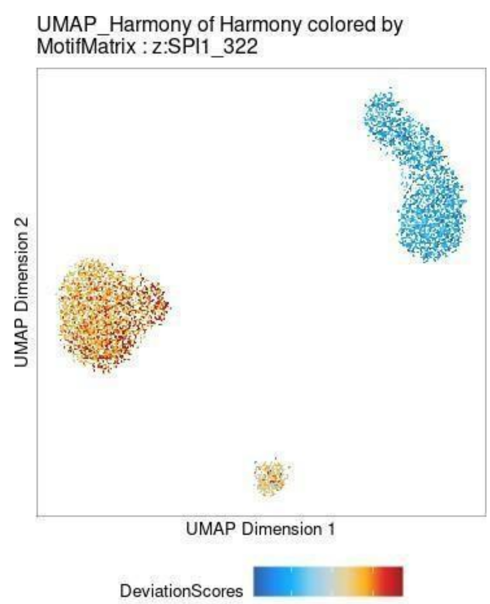


Figure 12. UMAP showing SPI1_322 motif activity.

6.3 Distribution of motif activity

The distribution of motif activity across clusters was examined for the top variable motifs.

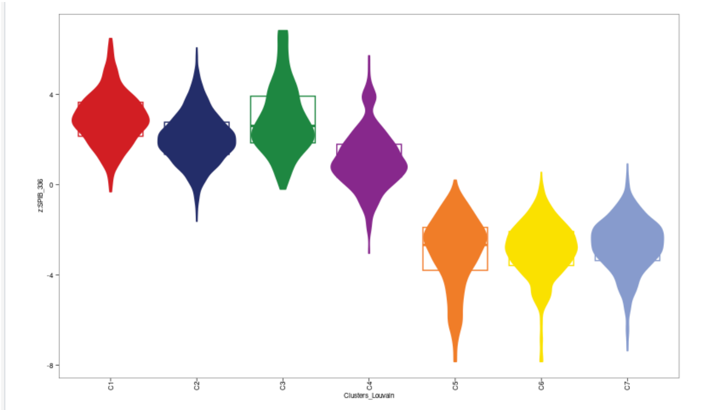


Figure 13. Distribution of SPIB_336 motif activity across clusters.

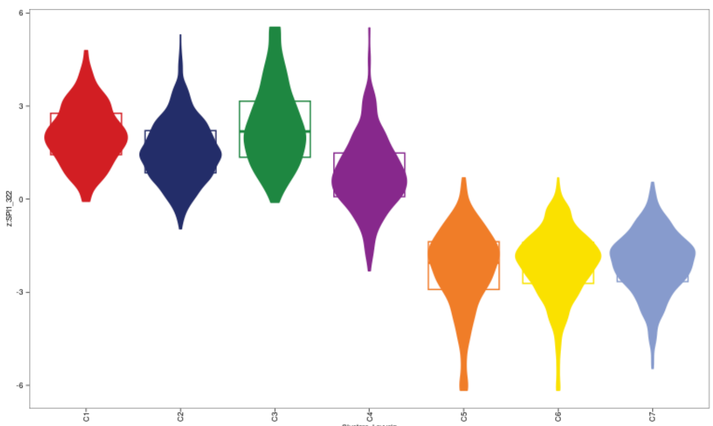


Figure 14. Distribution of SPI1_322 motif activity across clusters.

6.4 Interpretation of motif activity

A high motif activity score does not guarantee that the corresponding transcription-factor protein is present. chromVAR measures chromatin accessibility associated with TF binding motifs; it does not directly measure TF mRNA expression, protein abundance or direct TF occupancy.

SPIB_336 and SPI1_322 showed positive deviation scores in clusters C1–C4 and lower or negative scores in C5–C7. This indicates cluster-specific accessibility of the corresponding motifs rather than direct confirmation of TF expression.

6.5 PAX5 motif activity and gene expression

PAX5 motif activity was cross-referenced with PAX5 gene expression obtained from the GeneIntegrationMatrix. The PAX5 expression signal was concentrated in a distinct cluster, and PAX5 motif activity was enriched in the same cluster(s). The two measurements therefore largely agree.

Biologically, PAX5 is a lineage-defining transcription factor for B cells. Concordant accessibility at PAX5 motifs and PAX5 transcript expression is consistent with cell-type-specific regulatory activity.

7. Integration with scRNA-seq Gene Expression

7.1 Gene-expression integration

Annotated scRNA-seq data from the provided RDS file were integrated with the scATAC-seq dataset using ArchR's gene-integration framework. Gene expression was transferred to ATAC cells through the GeneIntegrationMatrix, allowing expression information to be projected onto the ATAC UMAP.

PAX5, EGR1 and OLIG2 were used as representative genes for visualization of gene-expression patterns in the scATAC-seq embedding.

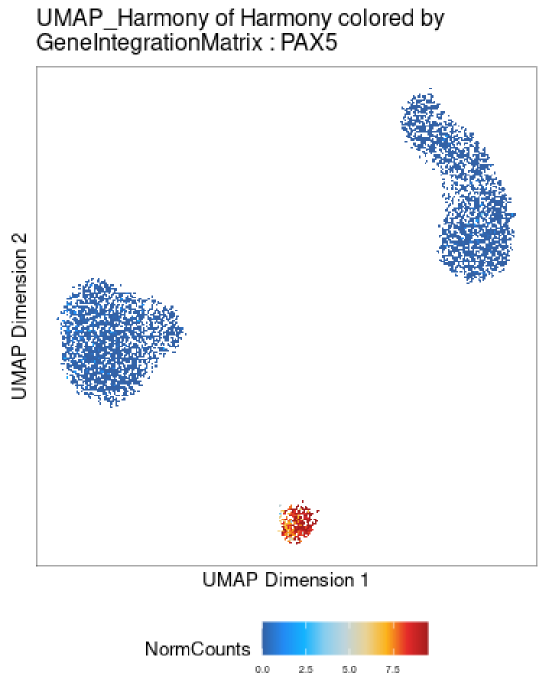


Figure 15. GeneIntegrationMatrix UMAP showing PAX5 expression.

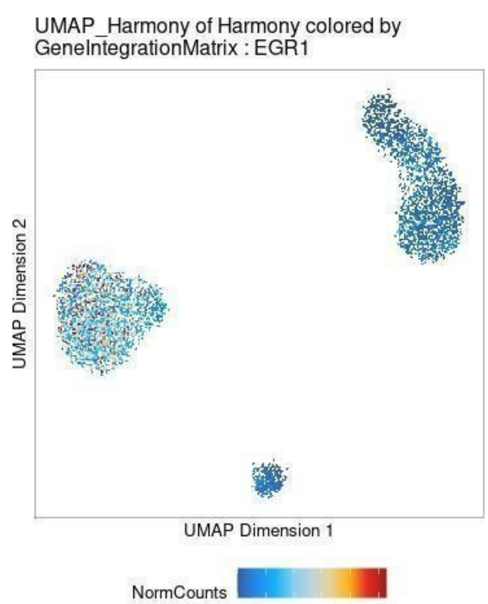


Figure 16. GeneIntegrationMatrix UMAP showing EGR1 expression.

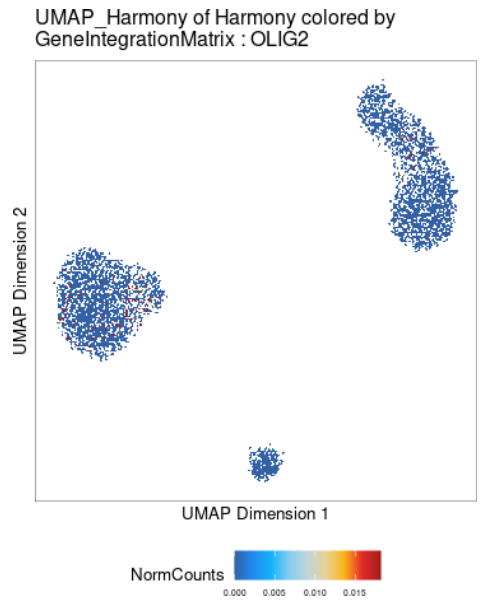


Figure 17. GeneIntegrationMatrix UMAP showing OLIG2 expression.

7.2 Correlation between gene activity and gene expression

Correlation coefficients were computed between scRNA-seq gene expression and scATAC-seq gene activity across the selected gene set. The overall distribution showed mostly weak to moderate positive correlations, indicating that chromatin accessibility is only partially predictive of gene expression.

The highest agreement was observed for DMXL2 (correlation = 0.96), indicating strong concordance between accessibility and transcription. The lowest agreement was observed for MALAT1 (correlation = −0.81), indicating poor correspondence between gene activity and gene expression for this gene.

7.3 Comparison of ATAC clusters with RNA-inferred cell types

RNA-derived cell-type labels were transferred to the scATAC-seq clusters, and a confusion matrix was generated to compare ATAC-derived clusters with RNA-inferred cell-type labels.

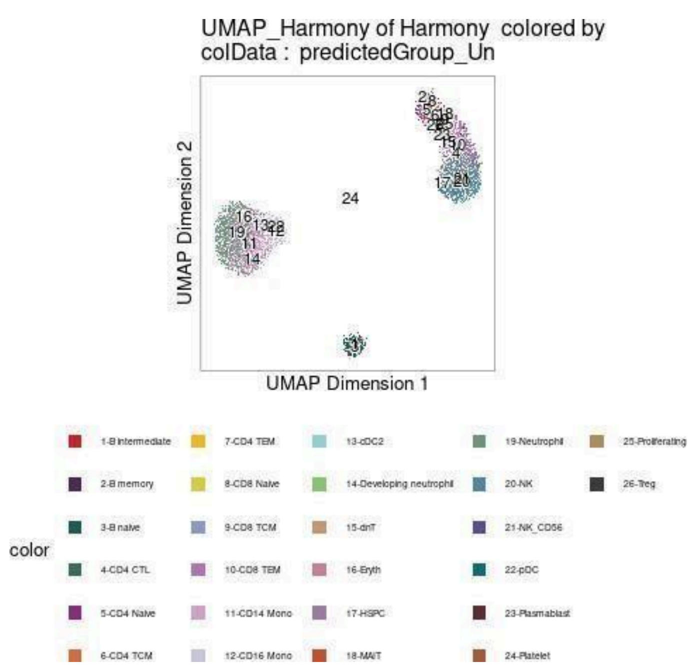


Figure 18. Comparison of ATAC clusters with RNA-inferred cell-type labels.

Several ATAC clusters showed strong agreement with specific RNA-defined cell types. For example, C2 was dominated by CD14 Monocytes and Neutrophils, C4 corresponded almost exclusively to B naïve cells, and C7 was strongly enriched for NK cells. Other clusters, including C5 and C6, contained multiple related immune subtypes, indicating partial mixing rather than a strict one-to-one correspondence. Overall, the agreement was good but not perfect, as expected because chromatin accessibility and gene expression measure different molecular layers and related immune populations can share regulatory programs.

7.4 Peak-gene linkage

Peak-gene linkage was computed by correlating chromatin accessibility at peaks with gene expression across cell types. The analysis identified thousands of candidate regulatory elements whose accessibility was associated with gene expression. Side-by-side heatmaps of peak accessibility and gene expression showed concordant, cluster-specific patterns, supporting cell-type-specific regulatory relationships.

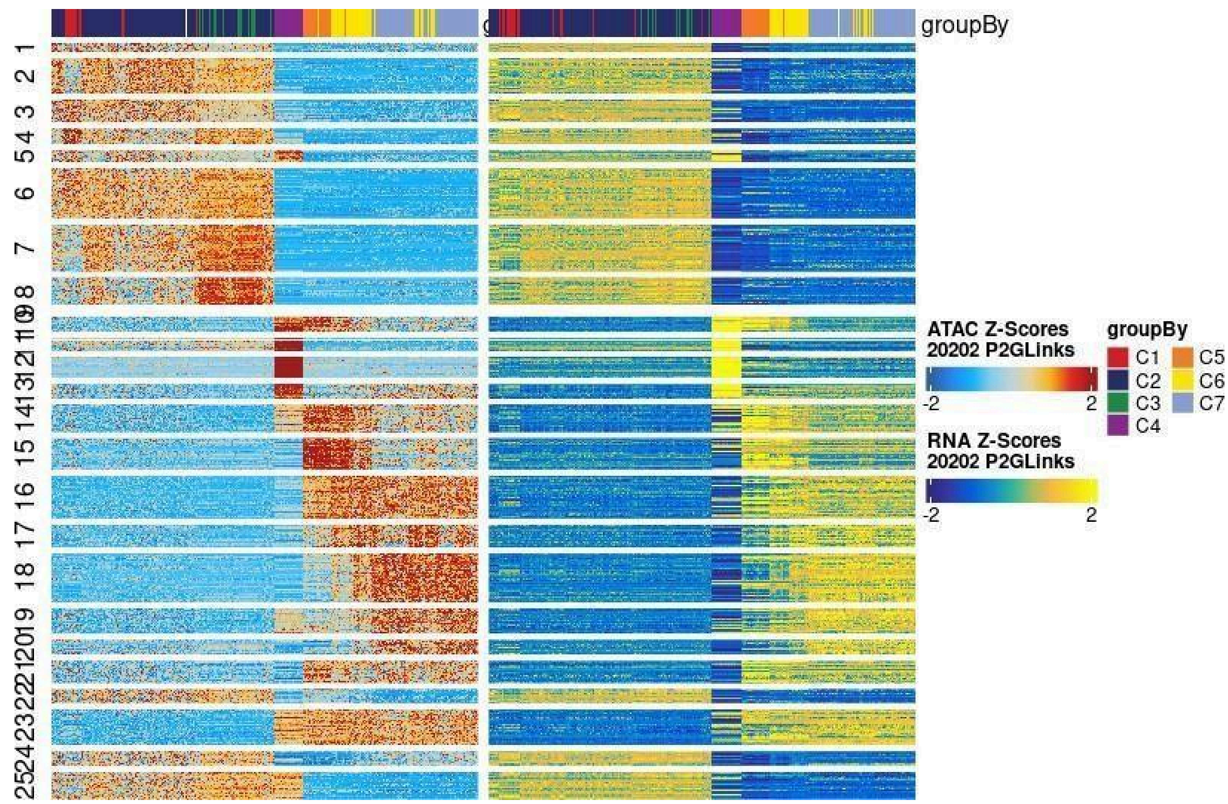


Figure 19. Side-by-side heatmaps of candidate peak accessibility and gene expression across cell types.

8. Differential Chromatin Accessibility

8.1 COVID-19 versus Healthy comparison

Differential peak accessibility was assessed between COVID and Healthy groups. MA and volcano plots were generated to examine the magnitude and statistical significance of accessibility differences.

dataset.

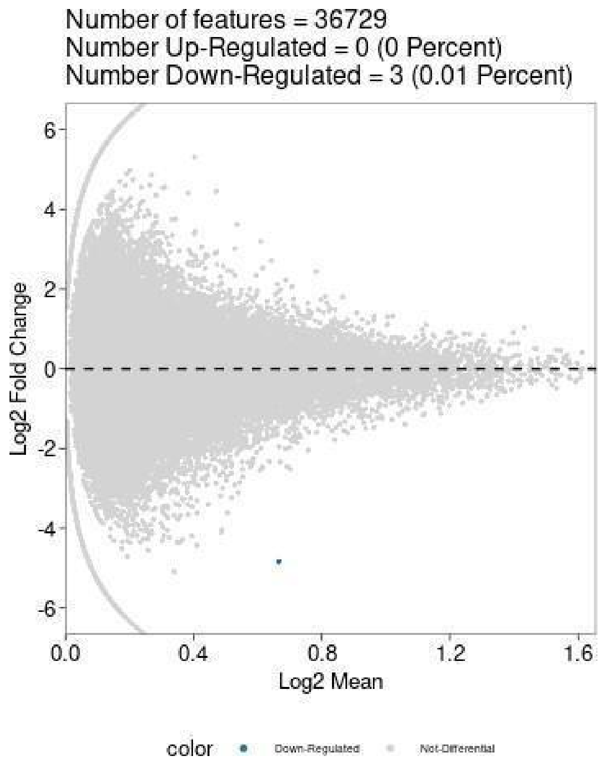


Figure 20. MA plot for differential peak accessibility between COVID and Healthy groups.

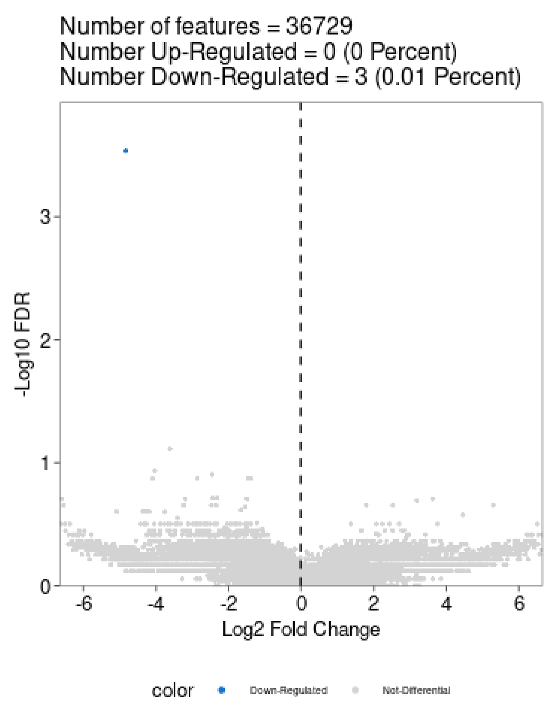


Figure 21. Volcano plot for differential peak accessibility between COVID and Healthy groups.

Most peaks were centred around log2 fold change ≈ 0, indicating similar chromatin accessibility between the two conditions. Only a very small number of peaks (approximately 2–3) showed statistically significant differential accessibility, and these were down-regulated in COVID. The results therefore indicate that global chromatin-accessibility patterns were largely conserved, with only subtle condition-specific differences in this dataset.

9. Transcription-Factor Motif Enrichment

9.1 Enrichment calculation and matched background

TF motif enrichment tests whether transcription-factor binding motifs are over-represented in a selected set of condition-associated peaks compared with a background peak set. In ArchR, peaks of interest are compared with background peaks matched for characteristics such as GC content and average accessibility, and statistical significance is assessed with multiple-testing correction (FDR).

A matched background is important because motif occurrence depends on sequence composition and chromatin accessibility. Comparing condition-specific peaks with all genomic peaks could introduce sequence and coverage bias. Matching the background helps ensure that observed motif enrichment reflects biological regulation rather than technical or compositional differences.

9.2 Healthy-specific motifs

No robust set of Healthy-specific peaks passed the significance thresholds in the differential accessibility analysis. Therefore, no TF motif families could be confidently identified as enriched in Healthy-specific peaks.

9.3 COVID-specific motifs

Similarly, no significantly up-regulated peaks were detected in COVID. Consequently, no TF motif families could be reliably reported as enriched in COVID-specific peaks.

9.4 TF motif activity visualization

TF motif activity was visualized using motif activity scores projected onto UMAP space and summarized across clusters. The visualizations did not show strong condition-specific separation, consistent with the limited differential peak accessibility detected between COVID and Healthy samples.

10. Transcription-Factor Footprinting

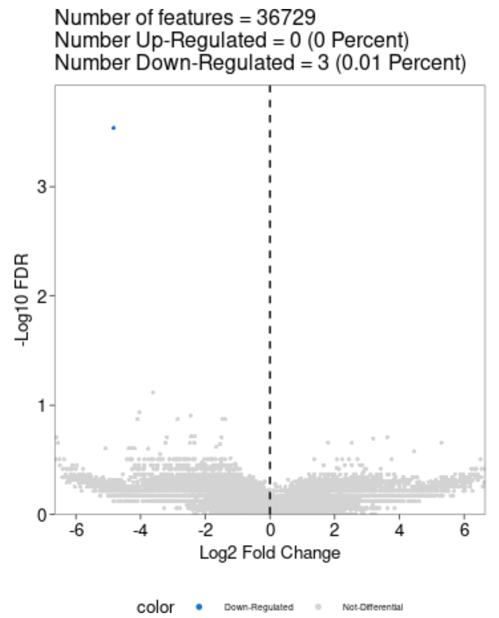
10.1 Footprinting on the condition-associated feature set

TF footprinting was attempted for the top motifs from the COVID versus Healthy comparison. Because no significantly enriched TF motifs were identified in the differential analysis, meaningful condition-specific footprinting for a top three motif set could not be performed.

10.2 Tn5 sequence bias correction

Tn5 transposase does not cleave DNA uniformly and has sequence-specific cleavage preferences. Without correction, apparent footprints can reflect sequence bias rather than true transcription-factor binding, producing false-positive or unreliable footprints.

Bias correction was performed using sequence-aware normalization, in which observed Tn5 insertion profiles are normalized against an expected cleavage profile derived from local DNA sequence context. This reduces sequence-driven insertion bias while preserving genuine chromatin-accessibility and TF-binding signals.



The sharp dip in the centre of a motif represents reduced Tn5 insertion caused by physical protection of DNA by a bound transcription factor. A deeper or more consistent dip is compatible with stronger TF occupancy.

The higher shoulders surrounding the motif represent increased Tn5 accessibility adjacent to the protected binding site. These flanking regions are accessible to Tn5 and are often nucleosome-free, producing the characteristic footprint shape.

11. Co-accessibility and Regulatory Elements

11.1 Co-accessibility of peaks

Co-accessibility between peaks was computed to identify genomic regions with coordinated chromatin accessibility. Genome-browser track plots were used to inspect regulatory relationships around marker genes.

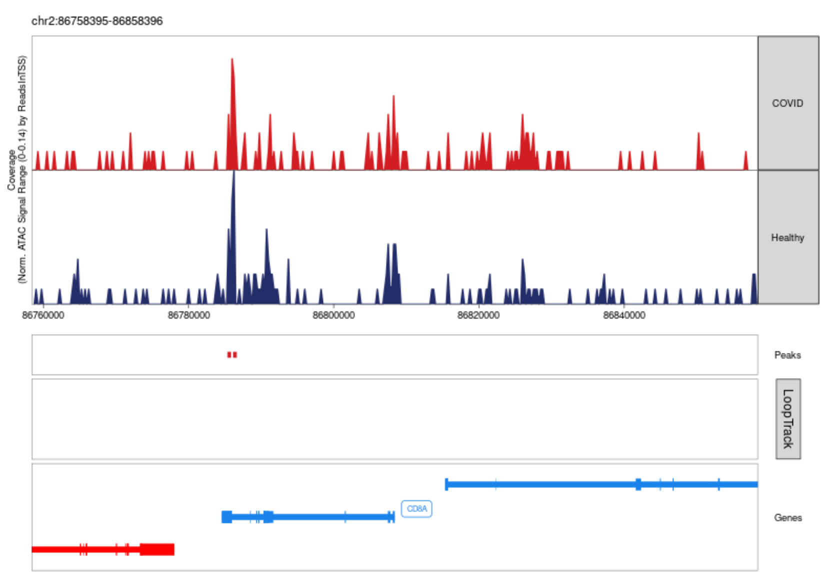


Figure 24. Co-accessibility and accessibility tracks around the CD8A locus comparing COVID and Healthy samples.

The accessibility tracks show multiple peaks across the CD8A locus, with differences in signal intensity between COVID and Healthy samples. Called ATAC-seq peaks overlap regions of high accessibility and therefore represent candidate regulatory regions.

Curved co-accessibility arcs connect distal peaks to the CD8A locus. The displayed interactions use a correlation cutoff of 0.5 and represent high-confidence coordinated accessibility between regulatory regions. The CD8A gene body and associated peaks indicate that regulation can involve multiple elements around the transcription start site rather than the promoter alone.

11.2 Potential enhancer identification

Potential enhancers were identified by examining accessible peaks and their co-accessibility/linkage to the transcription start site of highlighted marker genes. A distal accessible peak was counted as a potential enhancer when the analysis showed a corresponding regulatory linkage to the gene TSS under the applied criteria.

Marker gene	Potential enhancers
CD34	0
GATA1	0
PAX5	2
MS4A1	0
MME	0
CD14	2
MPO	0
IRF8	2
CD3	0
CD8A	0
CD4	1
TBX21	1
CD3G	0
NCAM1	0
FCGR3A	0
FOXP3	0
GATA3	0
RORC	0
PDCD1	0
HLA-DRA	0
CD28	1
IL2RA	0
CD69	3
CD44	5
CCR7	2

11.2.1 CD34

The CD34 locus showed a limited number of nearby accessible peaks but no strong distal co-accessible regions linked to the transcription start site, suggesting predominantly promoter-proximal regulation.

Potential enhancers identified for CD34: 0.

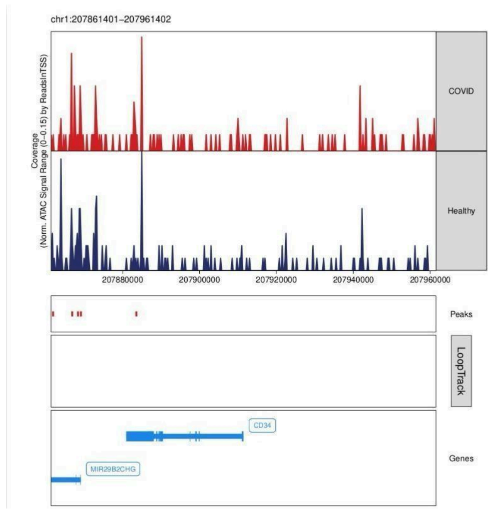


Figure 25.1. Genome-browser view of accessibility and regulatory links around CD34.

11.2.2 GATA1

GATA1 showed a strong promoter-associated accessibility peak in both conditions, with no distant co-accessible regions connecting to the transcription start site.

Potential enhancers identified for GATA1: 0.

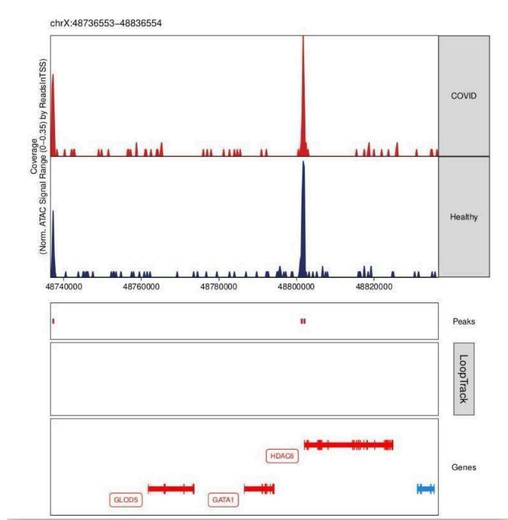


Figure 25.2. Genome-browser view of accessibility and regulatory links around GATA1.

11.2.3 PAX5

The PAX5 transcription start site was co-accessible with several distal accessible peaks, consistent with long-range regulatory interactions compatible with enhancer-mediated control.

Potential enhancers identified for PAX5: 2.

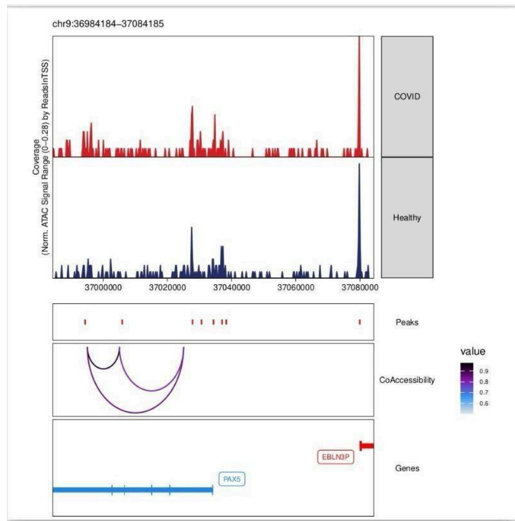


Figure 25.3. Genome-browser view of accessibility and regulatory links around PAX5.

11.2.4 MS4A1

MS4A1 showed robust promoter-associated accessibility but no distal co-accessible regions linked to its transcription start site.

Potential enhancers identified for MS4A1: 0.

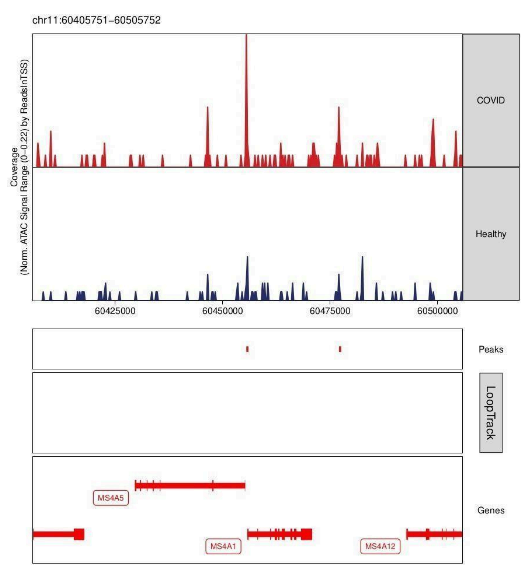


Figure 25.4. Genome-browser view of accessibility and regulatory links around MS4A1.

11.2.5 MME

MME showed a significant promoter-associated accessibility peak without discernible distal co-accessibility to the transcription start site.

Potential enhancers identified for MME: 0.

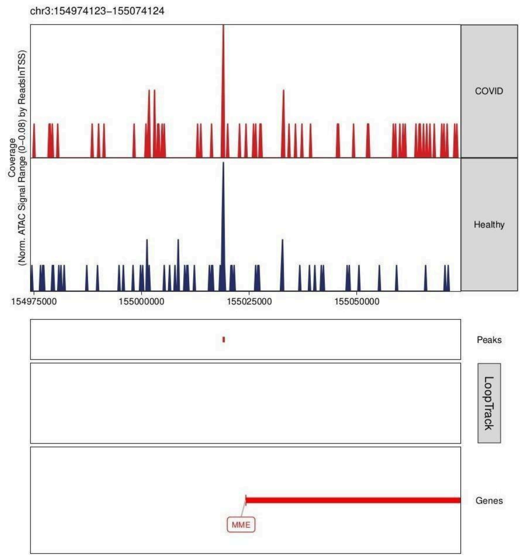


Figure 25.5. Genome-browser view of accessibility and regulatory links around MME.

11.2.6 CD14

CD14 showed many distal accessible peaks and significant promoter accessibility. Multiple co-accessibility linkages connected distal regions to the CD14 transcription start site, suggesting long-range regulatory interactions.

Potential enhancers identified for CD14: 2.

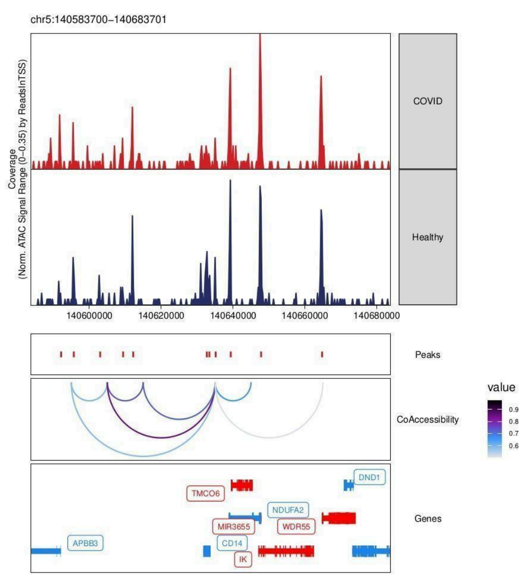


Figure 25.6. Genome-browser view of accessibility and regulatory links around CD14.

11.2.7 MPO

MPO showed strong promoter-proximal accessibility, especially in COVID, but no discernible co-accessibility connections to distant peaks.

Potential enhancers identified for MPO: 0.

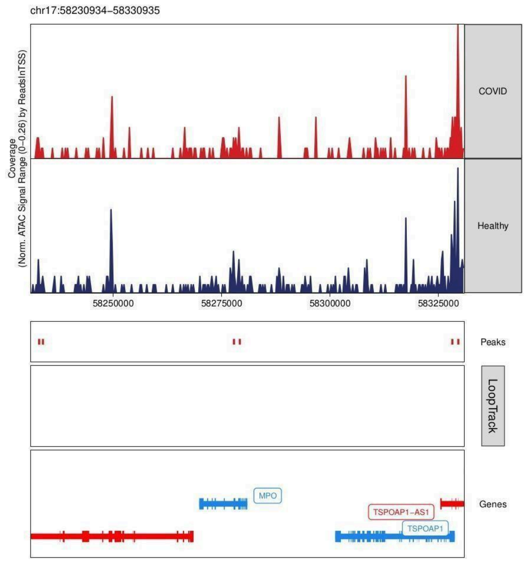


Figure 25.7. Genome-browser view of accessibility and regulatory links around MPO.

11.2.8 IRF8

IRF8 showed two co-accessibility linkages between the transcription start site and distal ATAC-seq peaks, suggesting distal regulatory elements that may function as enhancers.

Potential enhancers identified for IRF8: 2.

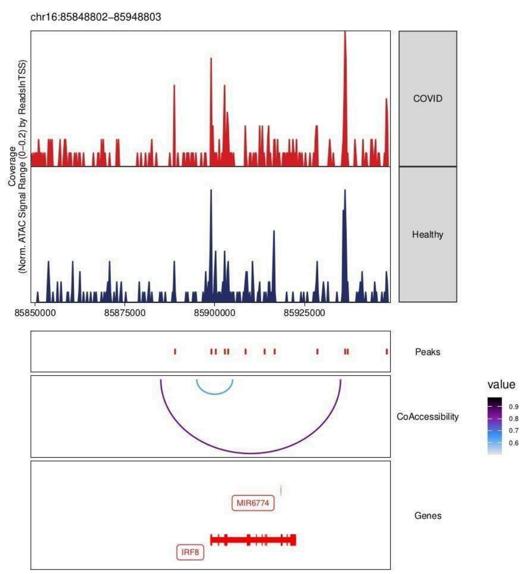


Figure 25.8. Genome-browser view of accessibility and regulatory links around IRF8.

11.2.9 CD3

Several accessible peaks were present around the CD3 region, but no discernible co-accessibility connections linked the CD3D/CD3G transcription start sites to distal peaks.

Potential enhancers identified for CD3: 0.

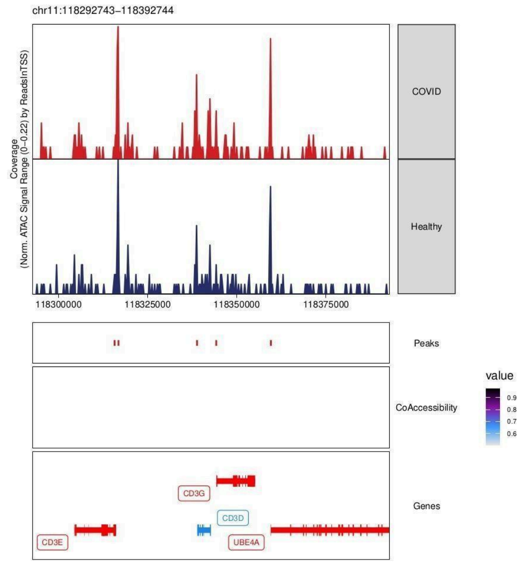


Figure 25.9. Genome-browser view of accessibility and regulatory links around CD3.

11.2.10 CD8A

CD8A showed strong promoter-associated accessibility in both conditions but no discernible distal co-accessibility connections under the applied criteria.

Potential enhancers identified for CD8A: 0.

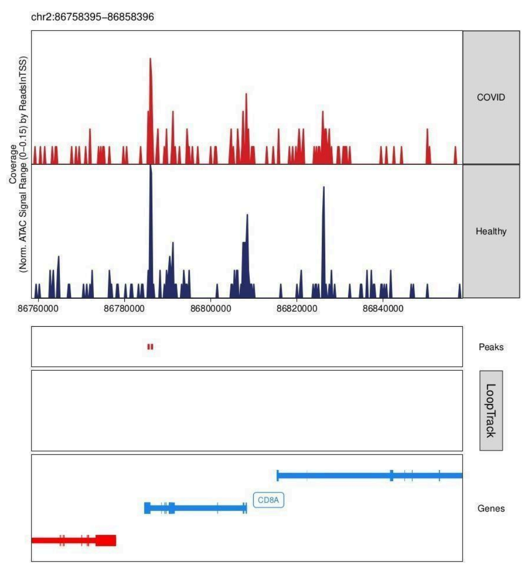


Figure 25.10. Genome-browser view of accessibility and regulatory links around CD8A.

11.2.11 CD4

CD4 showed high accessibility and several distal accessible peaks, including one with a strong co-accessibility connection to the transcription start site.

Potential enhancers identified for CD4: 1.

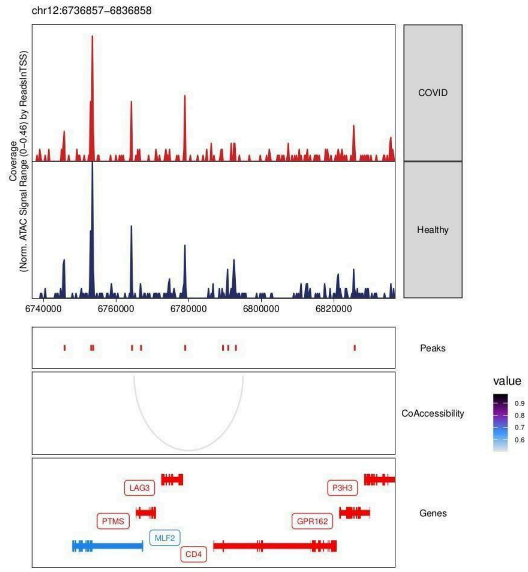


Figure 25.11. Genome-browser view of accessibility and regulatory links around CD4.

11.2.12 TBX21

TBX21 showed distinct accessibility in both conditions and a significant co-accessibility connection between a distal peak and the transcription start site, supporting a possible distal enhancer.

Potential enhancers identified for TBX21: 1.

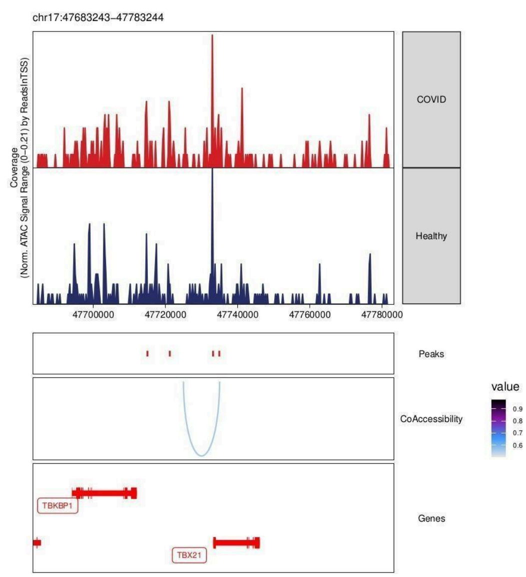


Figure 25.12. Genome-browser view of accessibility and regulatory links around TBX21.

11.2.13 CD3G

CD3G showed several accessible peaks but no discernible co-accessibility connections from the transcription start site to distal peaks.

Potential enhancers identified for CD3G: 0.

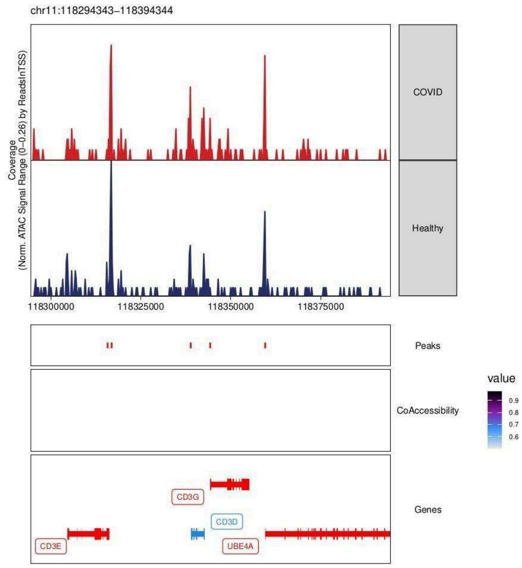


Figure 25.13. Genome-browser view of accessibility and regulatory links around CD3G.

11.2.14 NCAM1

NCAM1 showed accessible chromatin near the gene body but no strong distal peaks or co-accessibility connections to the transcription start site.

Potential enhancers identified for NCAM1: 0.

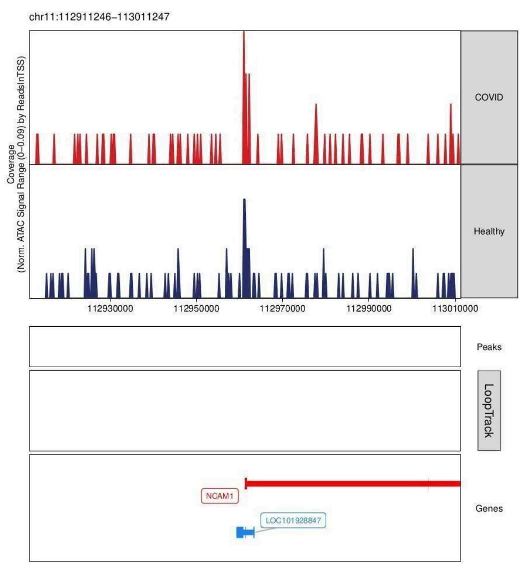


Figure 25.14. Genome-browser view of accessibility and regulatory links around NCAM1.

11.2.15 FCGR3A

FCGR3A showed accessible chromatin and multiple adjacent peaks but no discernible distal co-accessibility connections to the transcription start site.

Potential enhancers identified for FCGR3A: 0.

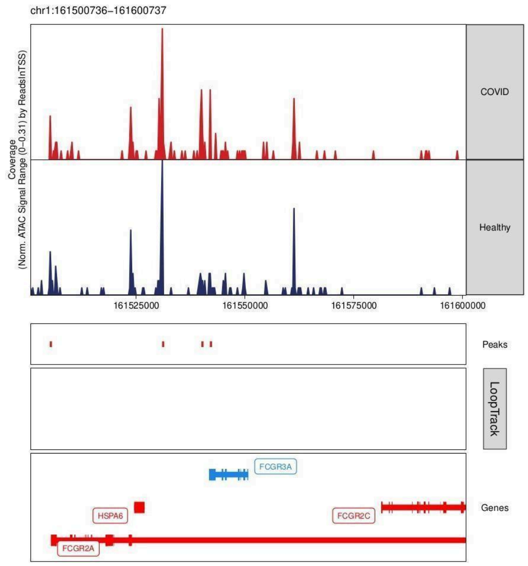


Figure 25.15. Genome-browser view of accessibility and regulatory links around FCGR3A.

11.2.16 FOXP3

FOXP3 showed a robust accessible promoter peak but no discernible co-accessibility connections to distal peaks.

Potential enhancers identified for FOXP3: 0.

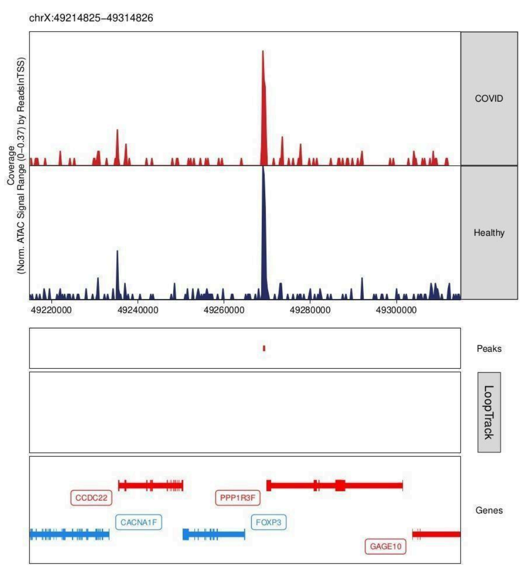


Figure 25.16. Genome-browser view of accessibility and regulatory links around FOXP3.

11.2.17 GATA3

GATA3 showed several accessible chromatin regions but no discernible co-accessibility connections between distal peaks and the transcription start site.

Potential enhancers identified for GATA3: 0.

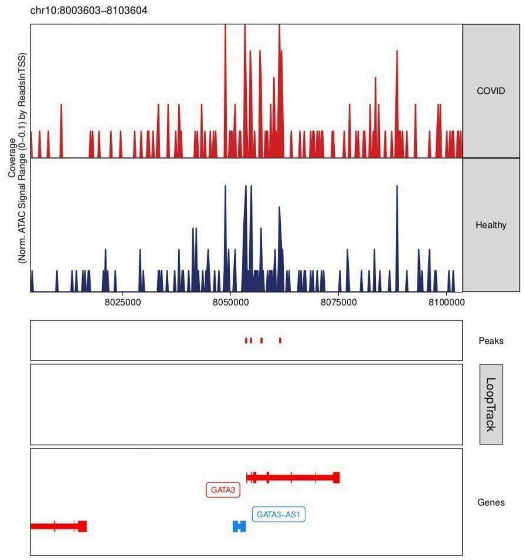


Figure 25.17. Genome-browser view of accessibility and regulatory links around GATA3.

11.2.18 RORC

RORC showed strong promoter accessibility and multiple nearby peaks, but no discernible distal co-accessibility connections to the transcription start site.

Potential enhancers identified for RORC: 0.

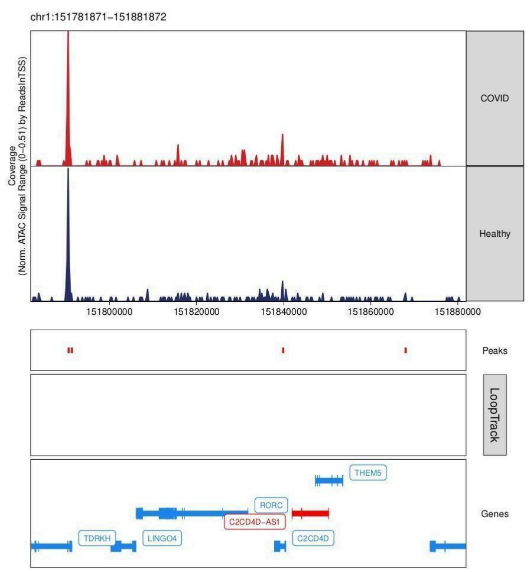


Figure 25.18. Genome-browser view of accessibility and regulatory links around RORC.

11.2.19 PDCD1

PDCD1 showed multiple distal accessible peaks and strong promoter accessibility, but no qualifying co-accessibility connections to the transcription start site.

Potential enhancers identified for PDCD1: 0.

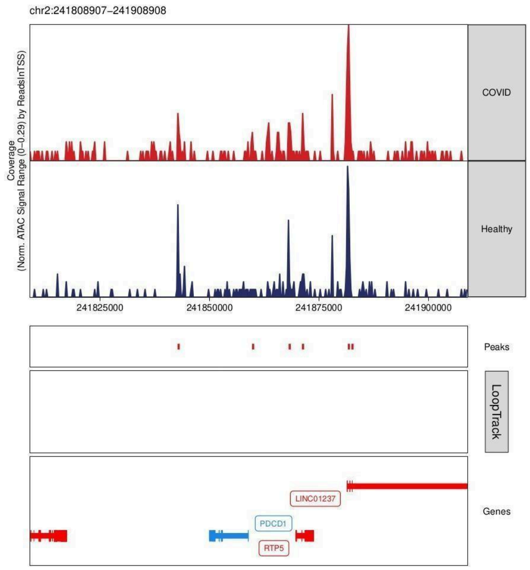


Figure 25.19. Genome-browser view of accessibility and regulatory links around PDCD1.

11.2.20 HLA-DRA

HLA-DRA showed strong promoter accessibility and multiple distal accessible peaks, but no discernible co-accessibility connections to the transcription start site.

Potential enhancers identified for HLA-DRA: 0.

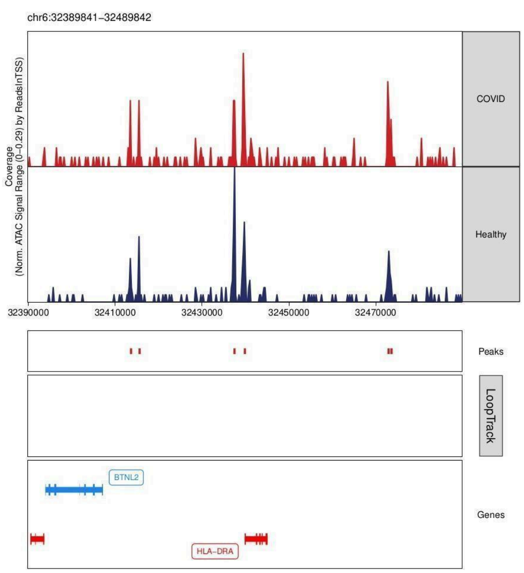


Figure 25.20. Genome-browser view of accessibility and regulatory links around HLA-DRA.

11.2.21 CD28

CD28 showed strong promoter accessibility and several distal peaks within the analyzed window. One distal peak showed substantial co-accessibility with the transcription start site, supporting a possible enhancer.

Potential enhancers identified for CD28: 1.

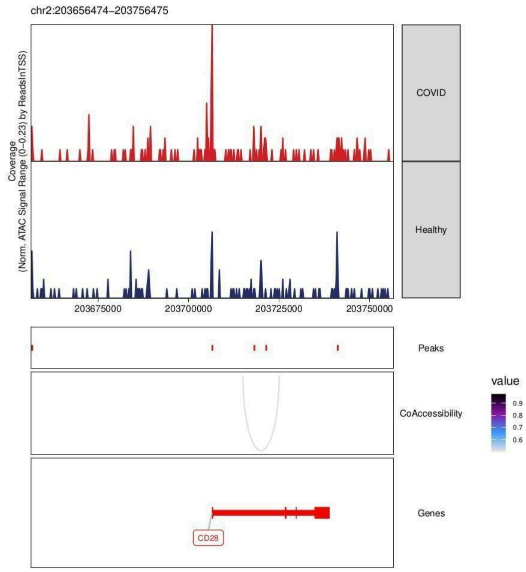


Figure 25.21. Genome-browser view of accessibility and regulatory links around CD28.

11.2.22 IL2RA

IL2RA showed an open promoter and several nearby accessible distal regions, but no noteworthy co-accessibility connections to the transcription start site.

Potential enhancers identified for IL2RA: 0.

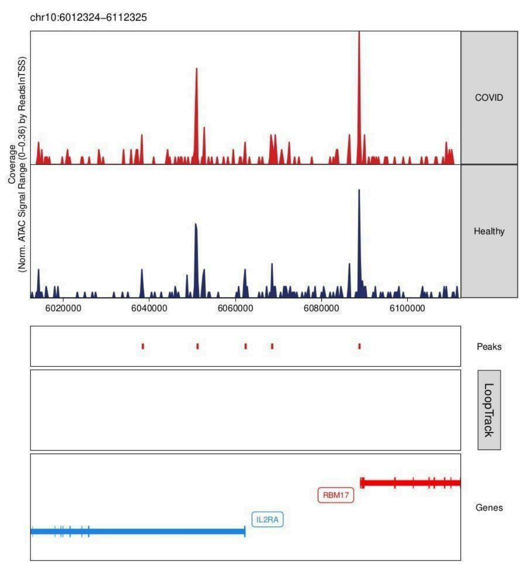


Figure 25.22. Genome-browser view of accessibility and regulatory links around IL2RA.

11.2.23 CD69

CD69 showed promoter accessibility and several distant accessible regions with significant co-accessibility to the transcription start site, suggesting multiple potential enhancers.

Potential enhancers identified for CD69: 3.

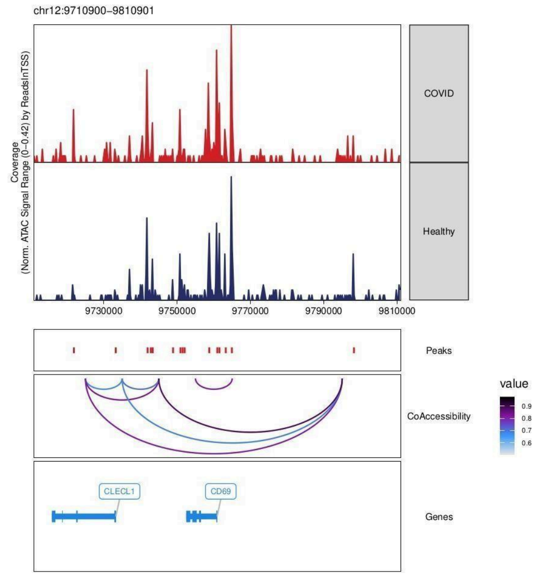


Figure 25.23. Genome-browser view of accessibility and regulatory links around CD69.

11.2.24 CD44

CD44 showed many distal accessible regions and significant promoter accessibility. Multiple co-accessibility connections to the transcription start site suggest several putative enhancers.

Potential enhancers identified for CD44: 5.

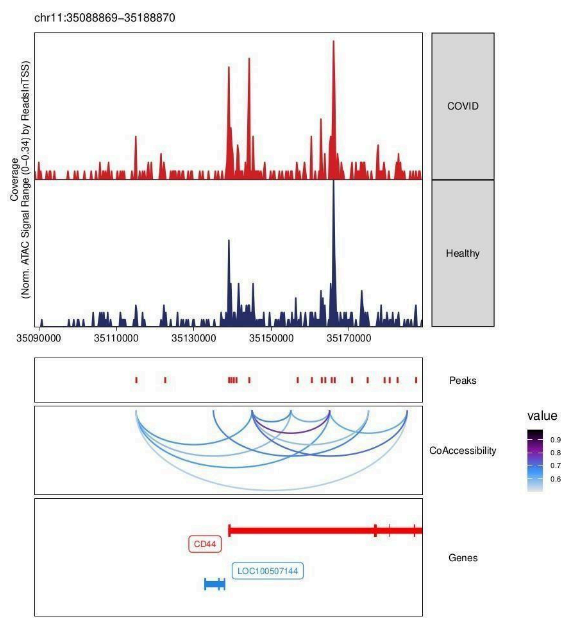


Figure 25.24. Genome-browser view of accessibility and regulatory links around CD44.

11.2.25 CCR7

CCR7 showed accessible promoter and distal regions, with two distal peaks showing strong co-accessibility with the promoter, suggesting putative enhancer elements.

Potential enhancers identified for CCR7: 2.

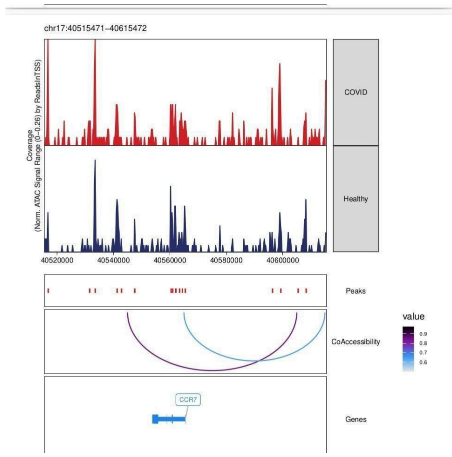


Figure 25.25. Genome-browser view of accessibility and regulatory links around CCR7.

12. Overall Findings

- Quality-control metrics were consistent across the four samples, with no clear low-quality outlier sample identified.
- After strict filtering and doublet removal, 5,089 cells were retained for downstream analysis.
- Pseudo-bulk peak calling and a joint PeakMatrix provided a suitable accessibility representation for differential and downstream analyses.
- Iterative LSI was used instead of PCA as the primary dimensionality-reduction approach because of the sparse structure of scATAC-seq data; LSI component 1 was excluded because it tracked sequencing depth.
- Harmony reduced sample-associated batch effects and improved mixing while preserving the biological structure of the UMAP.
- Seven Louvain clusters were identified in the final scATAC-seq dataset.
- Gene activity and cluster marker analysis identified cluster-specific regulatory patterns, and MAGIC was used to visualize smoothed gene-activity patterns.
- cisBP motif annotation and chromVAR deviation scores identified SPIB_336 and SPI1_322 as the most variable motifs among the motifs examined.
- Integration with scRNA-seq data enabled comparison of chromatin accessibility with gene expression and showed good but imperfect agreement between ATAC clusters and RNA-derived cell types.
- Differential accessibility between COVID and Healthy samples was limited, with most peaks near log2FC ≈ 0 and only a small number of significant changes.
- Because robust condition-specific motif enrichment was not detected, condition-specific TF footprinting could not provide a meaningful top-three-motif result.
- Co-accessibility analysis identified potential distal regulatory elements for selected genes, with the strongest numbers of candidate enhancers reported for CD44, CD69, PAX5, CD14, IRF8 and CCR7.

13. Reproducibility and References

The analysis was implemented in R using ArchR and associated single-cell chromatin-accessibility methods. The report preserves the analysis sequence, parameters, results and figures generated during the workflow.

Primary references used for the analysis:

- ArchR documentation: https://www.archrproject.com/bookdown/index.html
- Advanced single-cell RNA-seq/ATAC-seq analysis training material: https://ucdavis-bioinformatics-training.github.io/2022-July-Advanced-Topics-in-Single-Cell-RNA-Seq-ATAC/data_analysis/scATAC_analysis

- ArchR GitHub repository: https://github.com/GreenleafLab/ArchR